Climate Intelligence & Decision Twin for Dubai

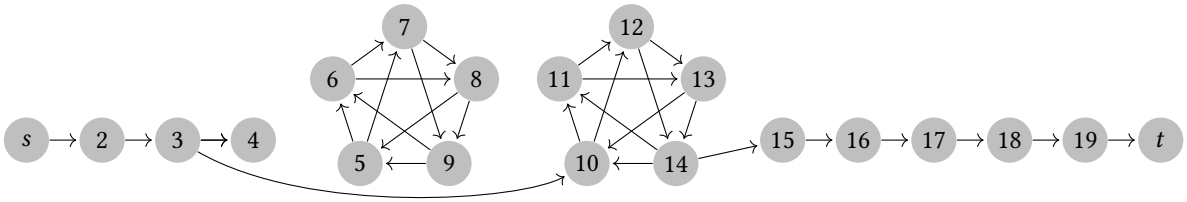
A full application to the *Dubai Future Foundation Research, Development and Innovation Grant* under the Cognitive Cities priority

The ask

Amount	[1,400,000.00:388,888.89](AED:USD)
Duration	36 months, 15 March 2027 to 14 March 2030
Phasing	Phase 1, 12 months; Phase 2, 24 months, released at a stop-or-go gate
Domain	Cognitive Cities: climate-neutral and resilient urban systems; urban health intelligence for proactive well-being

Applicants

Dr Chakradhar Iyyunni	Principal Investigator, BITS Pilani Dubai Campus, United Arab Emirates
Mahesh Shastry	Knowledge and systems architecture, Kalman Systems Pty Ltd, Sydney, Australia



Prepared on the rescoped basis approved at Letter of Intent: one instrumented site, named partners, named tools, funded people and one built intervention. The evidence base, the architecture and the costing method are those of the companion feasibility study [Shastry et al., 2026], whose source archive is registered in Appendix A (p. 16).

Private and confidential
For the funder and the endorsing institution

Front Matter

The Executive Summary and the epigraph. Who applies and who answers for the application; the Foreword and the Preface, which fix the evidence-grading scheme this document is bound by; what it assumes and what it declines to claim. Then the Contents, the figures, the tables, the acronyms and the symbols.

Executive Summary

Keywords: urban heat; decision twin; outdoor tourism capacity; minimum cooling network; pedestrian thermal comfort; physics-informed downscaling; semantic layer; knowledge graph; stochastic dynamic programming; life-cycle cost; Cognitive Cities

This application requests [1.4m:389k](AED:USD) over 36 months, 15 March 2027 to 14 March 2030, under the Cognitive Cities priority *[Dubai RDI Programme, 2025]*. It is released in two stages, twelve months and twenty-four, with a stop-or-go gate between them.

Dubai's outdoor economy runs on hours that heat takes away, and the emirate's climate data describes the heat rather than what the heat prevents. The region's only measured urban-climate study validated a street model against a 26-station network to 0.27–1.62°C after bias correction, then stopped at temperature (measured) [Manandhar et al., 2019]; Destination Earth resolves 500–750 m regionally (method-reference) *[ECMWF, 2024]*, orders of magnitude above the pavement where the loss happens. Every element of an answer exists somewhere and the intersection is empty: nothing ties activity-hours lost to heat, at corridor scale, to a built intervention tested against a prediction registered before construction.

Three constructs close that gap. Outdoor Tourism Capacity (OTC) counts how many visitors can comfortably use a place, for how long and at which hours; Heat-Lost Activity Hours (HLAH) count what heat removes. The Minimum Cooling Network (MCN) is the smallest set of shade, planting, surface and rest-point interventions whose Restored Activity Hours (RAH) meet a stated threshold on every priority route, costed across capital, energy, water and maintenance. The Decision Twin compares those options before anything is built and reports the comparison as a brief of two pages, not as a three-dimensional rendering; it is not a competing twin but the hour-by-hour comfort layer that the Dubai Digital Twin Platform *[Dubai Media Office, 2026]* does not carry, which is why the scope fits this envelope rather than one an order of magnitude larger.

About one kilometre of Dubai Creek waterfront is the testbed. The passively cooled heritage core at Al Fahidi is the untreated control across three summers, the 800 m of open road from Al Ghubaiba metro station to it is the intervention site, and nine instrumented nodes, two of them reference-grade, record both zones every ten minutes. The third year builds one structure and measures it against the prediction the twin lodged beforehand, so that the twin can be found wrong on the record.

One published finding already shapes the design. In Phoenix, reflective coating left streets about 12.0°F cooler at the surface at noon than adjacent aged asphalt, yet raised mean radiant temperature at the road centre by 5.8°F, demanding a further 12–13 watts per square metre of a standing person (measured-with-controls) [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]. Shade therefore ranks ahead of coatings, and the optimiser scores mean radiant temperature rather than air temperature.

The month-twelve gate requires four conditions together: 90 per cent sensor uptime across the summer; street-model error within 1.0°C on bias-corrected hourly air temperature; an hour-by-hour comfort map of both zones; and written confirmation from the Roads and Transport Authority or the Dubai Electricity and Water Authority that the output is useful for a decision it faces. The fourth is another organisation's judgement and the one condition the consortium cannot satisfy by working harder; it is carried as a risk and not as an assumption. A failed gate closes the project with the dataset and the platform as its deliverables.

The decision requested Approval of [1.4m:389k](AED:USD) over 36 months on the two-stage basis set out here, the second stage released only against the four gate conditions.

Your children are not your children.
They are the sons and daughters of Life's longing for itself.
They come through you but not from you,
And though they are with you yet they belong not to you.

You may give them your love but not your thoughts,
For they have their own thoughts.

You may house their bodies but not their souls,
For their souls dwell in the house of tomorrow,
which you cannot visit, not even in your dreams.

Kahlil Gibran, *The Prophet*, 1923

أولادكم ليسوا لكم
أولادكم أبناء الحياة المشتاقة إلى نفسها
بكم يأتون إلى العالم، ولكن ليس منكم
ومع أنهم يعيشون معكم، فهم ليسوا ملكاً لكم

أنتم تستطيعون أن تمنحوهم محبتكم
ولكنكم لا تقدرون أن تغرسوا فيهم بذور أفكاركم
لأن لهم أفكاراً خاصة بهم

وفي طاقتكم أن تصنعوا المساكن لأجسادكم
ولكن نفوسهم لا تقطن في مساكنكم
فهي تقطن في مسكن الغد

جبران خليل جبران

Qur'an Manuscript Folio, 1500s

Afghanistan, Herat, Safavid period (1501–1722)



Calligraphy, the art of beautiful writing, was elevated above all other art forms in the Islamic world because Allah (God) revealed the divine word of Islam to the Prophet Muhammad (570–632) in the Arabic language. This beautiful double page forms the opening pages, or *unwan*, of a Qur'an, the sacred book of Islam. Read from right to left, the verses are written in Arabic in elegant *naskhi* script on a gold ground with florets identifying the ends of the verses. Calligraphers who specialized in beautiful writing often dedicated their lives to copying the Qur'an to grow closer to Allah and receive his blessings. The pages are enhanced with splendid illumination: ornamentation in colors and gold with scrolling vines, blossoms, lozenges, and cartouches within bordered rectangles. Because the book arts were held in high esteem in the Islamic world, decorative motifs created by illuminators were often adopted in other art forms such as metalwork, textiles, and carpets.

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Applicants

Dr Chakradhar Iyyunni

Principal Investigator. Professor of Practice, BITS Pilani Dubai Campus, United Arab Emirates

Thirty years and more across academia and industry, working at the intersection of engineering, sustainability and systems innovation. His work at BITS Pilani Dubai Campus covers digital twins and automation, industry collaboration for energy transformation, and sustainable mobility, infrastructure and regenerative cities.

He is a Fellow at the European School of Governance and an Advisory Council Member at Ecocivilisation, India. Earlier roles with L&T Technology Services, the University of Houston and SmithKline Beecham span engineering consulting, leadership in research and development, and academic programme design, and carry a commitment to systems thinking and human-centred development.

He holds a PhD in a field relevant to the priority areas and is fully affiliated with the Dubai higher education institution named above, which are two of the three conditions the call sets for a Principal Investigator *[Dubai RDI Programme, 2025]*; the digital-twin and sustainable-mobility record described here is the third. The endorsing institution certifies all three.

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Mahesh Shastry

Knowledge and systems architecture. Kalman Systems Pty Ltd, Sydney, New South Wales, Australia

Interdisciplinary knowledge across science, engineering, industry and business, and wide-ranging experience that equips him to read the landscapes already in place. He brings to the work clarity, considered thinking and simplicity.

He is an advocate for the NSPE Code of Ethics for Engineers and works to the standards of ISO, IEC, IEEE-SA and BSI. He is a member of Engineers Australia, the Institute of Electrical and Electronics Engineers, the Institution of Engineering and Technology, the Association for Computing Machinery, ASM International, the American Mathematical Society and SPIE.

He is responsible here for the knowledge and systems architecture: the semantic layer that fixes what a place, an hour, a cohort and a lost hour mean before a sensor is bought, and the evidence discipline under which every quantity in this application carries a source and a grade.

Kalman Systems Pty Ltd, ACN 660 552 670.

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Collaborating entity. Curtin University Dubai joins as the collaborating entity, responsible for Green AI, machine-learning downscaling and the multi-objective optimisation behind the Minimum Cooling Network (MCN). It holds a budget line of its own, non-zero and carried in the second and third project years, rather than an acknowledgement in the text [Iyyunni, 2026]. The lead institution and the collaborating entity both sit in Dubai and

each does work the other cannot; the architecture comes from Sydney. A signed budget share and a Letter of Collaboration formalise the arrangement before the grant begins on 15 March 2027.

Investigators and funded posts. Four co-investigators at the lead institution carry mobility, the built environment, edge sensing and systems integration; they are named at award, with their doctorates and affiliations evidenced then, rather than in advance. Two posts are funded: a doctoral student in exposure and behaviour analysis from the first month of the grant, and a postdoctoral researcher in climate modelling and validation from the third, after procurement, both running to the thirty-sixth (design-target) [Iyyunni, 2026]. The institutional support the call requires of the endorsing institution, the reduced teaching load included, is undertaken in its Letter of Institutional Commitment *[Dubai RDI Programme, 2025]*[1].

Platform partner. No platform is contracted and none is budgeted in the first year. The Decision Twin interface is built in house in the second year, which makes a commercial platform partner an option to be exercised after the month-twelve gate and not a dependency; any such contract is placed after award [Iyyunni, 2026].

'A society grows great when old men plant trees in whose shade they shall never sit.'

Greek proverb

Circulated in English as a Greek proverb. No classical text carrying the wording has been identified, and no named author can be given for it; it is quoted here as a proverb and not as a citation.

[1] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

Acknowledgements

This application rests on documents its authors did not write and on data held by agencies that have not been asked to endorse it. The Dubai Future Foundation sets the envelope, the priority areas and the ten criteria against which a full application is scored, through its Research, Development and Innovation grant programme *[Dubai RDI Programme, 2025]*[1] *[Dubai Future Foundation, 2025]*[2]. Four Dubai bodies hold what the Al Fahidi and Al Seef testbed depends on: one named dataset each from the Roads and Transport Authority and the Dubai Electricity and Water Authority, and one written site permission each from Dubai Municipality and Dubai Culture [Iyyunni, 2026]. The National Center of Meteorology holds the wider climate record. The Dubai Digital Twin Platform, launched on 2 July 2026 (grade: policy-fact) *[Dubai Media Office, 2026]*[3], is the surface the Decision Twin sits on rather than competes with. None of these bodies has endorsed this application, not one of the four letters has been signed, and nothing here should be read as their view.

The field evidence comes from work done elsewhere, and this application is bound by what those authors actually published. Arizona State University and the City of Phoenix measured coated pavement against uncoated pavement on the same day, and what they found at the height of a standing person orders the candidate interventions here [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]. The Ville de Paris, through the Urban Innovative Actions initiative, published the OASIS schoolyard protocol from which the paired-control design is taken, and published no effect sizes; it is used for its design and nowhere for a result [Urban Innovative Actions, 2023, OASIS research team, 2024a]. Cooling Singapore supplies the nearest comparable urban climate twin, and with it the collision over OTC, which is Outdoor Tourism Capacity here and outdoor thermal comfort in their literature [Cooling Singapore 2.0, 2024]*[Centre for Liveable Cities, 2024]*[4]. The International Labour Organization and the Lancet Countdown on Health and Climate Change supply the labour and exposure series that HLAH must reconcile with [ILO, 2019, Romanello et al., 2024]. The Abu Dhabi local climate zone study is the one UAE precedent for the error band admitted at the month-twelve gate [Manandhar et al., 2019].

Every quantity printed here carries one of seven evidence grades: measured-with-controls, measured, modelled, projection, policy-fact, case-report or method-reference.

[2] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

[3] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

[4] Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

A figure takes the grade of its weakest link; arithmetic performed for this application is shown and labelled derived or indicative; and a figure that states an intention rather than an estimate is marked as a design-target. The grades are a judgement about evidence and not a comment on the care of the original authors, several of whom state the limits of their own work more candidly than a register can. Where a source will not bear what a reader might wish, that is said plainly. Appendix A (p. 16) records the grade each source carries, the two named but not reviewed, and the defects that make particular numbers unsafe to quote; the companion study holds the dated archive behind it [Shastry et al., 2026].

Responsibility for what remains, for every derived figure, every indicative assumption and every error, rests with the applicants alone.

28 September 2026

Foreword

We are not asking the Dubai Future Foundation to fund another map of how hot Dubai is. Maps of that kind exist at several resolutions, and we cannot name one procurement decision they have changed. The question we want funded is the one that comes after them, and that nobody has yet answered for this city: which activities, routes and places are lost to heat, for how many hours of the year, for which people, and which single intervention restores the most of them per dirham, per litre of water and per kilowatt-hour of cooling load [Iyyunni, 2026]. Those three denominators are the whole of our ambition. They turn a climate product into a procurement instrument. They are also why we propose one instrumented kilometre of Dubai Creek rather than a citywide promise: [1.4m:389k](AED:USD) over 36 months buys twenty microclimate nodes, two summers of street-scale simulation, one intervention chosen against that arithmetic and one honest before-and-after measurement against an untreated control. What it does not buy is the structure itself: the Year-3 intervention is built with a partner, under written permission from the municipal and heritage authorities, and neither its construction nor its upkeep is charged to this grant [Iyyunni, 2026]. Nor does it buy a city, and we have kept the indoor extension we would like to build, Building Climate and Infection Intelligence, outside this application entirely.

Why this project rather than another? Because the published record contains every element of our claim and not their intersection. Abu Dhabi has shown that a street model can be brought within a root-mean-square error of 0.27–1.62°C of a 26-station network after empirical bias correction, and that a model calibrated on one local climate zone predicts another of the same class within 0.24–0.68°C (grade: measured) [Manandhar et al., 2019]; the first of those ranges, and not a figure of our own choosing, is where we set the gate's error threshold. Paris has published a paired-control schoolyard design good enough to copy [OASIS research team, 2024a, Urban Innovative Actions, 2023]. Phoenix has measured coated pavement against uncoated pavement on the same day [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]. Singapore has built the urban climate twin [Aydt et al., 2026, Cooling Singapore 2.0, 2024]. Destination Earth reaches 500–750 m regionally (grade: method-reference) *[ECMWF, 2024]*[5], and the Dubai Digital Twin Platform carries more than 195,000 buildings across more than 1,500 geospatial layers, with rainfall as its one named simulation (grade: policy-fact) *[Dubai Media Office, 2026]*[3]. Not one of them reports activity-hours lost to heat at corridor scale, tied to an intervention that was actually built, and checked against a prediction made before it was built. That intersection is the contribution, and the Decision Twin is designed to sit on the city's platform rather than beside it.

[5] European Centre for Medium-Range Weather Forecasts, *Fact sheet: ECMWF and the EU's Destination Earth initiative*, ECMWF media centre, 2024. https://www.ecmwf.int/en/about/media-centre/focus/2024/fact-sheet-ecmwf-and-eus-destination-earth-initiative

We should be plain about the artificial intelligence in this application, because the phrase is carrying too much weight in too many proposals. We are not placing a model on a climate dataset. We are building a chain: physical system, data layer, semantic layer, knowledge graph, reasoning and prediction, decision twin, action, learning. Its middle links are six different things, with six owners and six failure modes, and we have written them as six deliverables because they fail separately. A data model fails by losing provenance [W3C, 2013]; a semantic model fails when two agencies mean different things by 'place' or 'hour'; a knowledge graph fails by being unqueryable; a prediction model fails by error; a digital twin fails by becoming a rendering; a decision system fails by producing briefs that nobody acts on. Listing them separately is what allows a reviewer, and later an auditor, to see which one broke.

The last of those failures is the one that ends the project, so we have made it a gate rather than a hope. The fourth condition at month twelve requires at least one of the Roads and Transport Authority or the Dubai Electricity and Water Authority to confirm in writing that the comfort map is useful for a decision it actually faces [Iyyunni, 2026]. We would rather be stopped in March 2028, with the dataset and the platform delivered, than hold 24 further months of public money for a product no operational agency has asked for. The first cycle of this programme funded 24 of 374 proposals, about one in sixteen (derived) *[Dubai Future Foundation, 2025]*[2]; a proposal that cannot say under what conditions it should be closed has not earned that ratio.

Two disciplines govern this document, and both have cost us something. The first is evidence grading: every figure carries one of seven grades, from measured-with-controls to method-reference, and takes the grade of its weakest link. We had meant to argue for reflective coatings from the Phoenix finding that treated residential streets run about 12.0°F cooler at noon than adjacent aged asphalt (grade: measured-with-controls). The same reports record mean radiant temperature at the road centre about 5.5°F *higher* at noon and in the afternoon, and find that a standing person needs 12–13 W m^{-2} more heat loss (grade: measured-with-controls) [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]. The surface is cooler and the pedestrian is not. So shade leads our candidate set, coatings are conditional, and the optimiser carries mean radiant temperature and not air temperature. Paris cost us a result as well: we wanted its effect sizes and found only its design. A figure that states an intention rather than an estimate is marked a design-target, including where it appears in the storyboard from which this application grew.

The second discipline is vocabulary: each key concept is set out in plain words, in professional terms and in technical terms, with a model where one exists [Beck et al., 2002]. Its price is length and candour. It obliged us to record that OTC means Outdoor Tourism Capacity here and Outdoor Thermal Comfort in the Singapore literature, and that our nodes as first specified could not measure the comfort index we intend to publish, which is why a black-globe thermometer now appears in the equipment line [ISO, 2017b, Jendritzky et al., 2012].

Every source is registered in Appendix A (p. 16) with its grade and its defects, and so are the sources we could not obtain [Shastry et al., 2026]. We ask the reviewer to prefer the source to this application wherever the two disagree.

Dr Chakradhar Iyyunni
Principal Investigator, BITS Pilani Dubai Campus
September 2026

Preface

This is a full application to the Dubai Future Foundation Research, Development and Innovation Grant under the Cognitive Cities priority. It asks [1.4m:389k](AED:USD) over 36 months, 15 March 2027 to 14 March 2030, to instrument one kilometre of Dubai Creek, model it at street scale, build one cooling intervention on it and measure what that intervention restores. It keeps the rescoped basis on which the Letter of Intent was written: one site, named partners, named tools, funded people and one built intervention [Iyyunni, 2026]. Its evidence base, its costing method and the conventions below are those of the companion feasibility study, which tested this proposal against the public record and is cited here wherever a figure comes from it [Shastry et al., 2026].

The genre and what it must contain

A full application is neither a research paper nor a prospectus. It is a document scored against ten published criteria by reviewers who must find each of them, and the main matter is therefore ten sections long, one section to a criterion, in ten pages. The rules that bind the ask are stated once here (grade: policy-fact): awards of [500k:139k](AED:USD) to [1.5m:417k](AED:USD) per project; an annual budget per project that must not exceed [500k:139k](AED:USD); a duration of one to three years; a Principal Investigator fully affiliated with a Dubai-based higher education institution, research centre or institute, holding a PhD in a field relevant to the priority areas and able to show a track record of research excellence; and at least 50 per cent of the grant utilised within Dubai *[Dubai RDI Programme, 2025]*[1]. The ask averages [467k:130k](AED:USD) a year (derived: [1.4m:389k](AED:USD) divided by three), but an average year is not a year and the cap is a test each year must pass on its own: Section 2.5 (p. 12) sets out the three annual totals, none of which exceeds the cap and the largest of which, Year 1, meets it exactly. The first cycle funded 24 of 374 proposals, about one in sixteen (policy-fact for the two counts, derived for the ratio) *[Dubai Future Foundation, 2025]*[2]. At those odds, a document that makes a reviewer hunt for a criterion is competing with a handicap it chose for itself.

Audience

The application is written first for the reviewer holding the rubric, who should be able to score each criterion from one section without reading the rest. It is written second for the endorsing institution, BITS Pilani Dubai Campus, which must demonstrate the institutional support measures the call names: a reduced teaching load for the faculty member leading the project, fast-tracked procurement, expedited hiring, flexibility to recruit research staff including assistants and engineers, a Letter of Institutional Commitment and a Letter of Collaboration *[Dubai RDI Programme, 2025]*[1]. It is written third for the collaborating entity, Curtin University Dubai, and for the four Dubai bodies on

whose single named datasets and written permissions the programme depends: the Roads and Transport Authority, the Dubai Electricity and Water Authority, Dubai Municipality and Dubai Culture. Every figure can be followed to its source through Appendix A (p. 16).

From the funder's criteria to the sections

Table 1 (p. xii) lists the ten full-application criteria in the funder's own order and published wording, each against the section that answers it *[Dubai RDI Programme, 2025]*[1]. Five criteria governed the Letter of Intent (relevance to the priority areas, partnerships and capacity building, innovation in methodology and outcomes, potential for impact in strategic sectors, and budget adequacy), and each is carried forward here, into Sections 1.4 (p. 6), 2.1 (p. 8), 1.2 (p. 3), 2.6 (p. 13) and 2.5 (p. 12) respectively, so that a reviewer who saw the letter can check the answer against what was promised.

How the evidence is graded

Every number, date and claim carries one of seven evidence grades. *Measured-with-controls*: an effect measured in the field against a simultaneous or paired control, such as a treated street beside an untreated one on the same day. *Measured*: quantities observed with instruments, without an intervention or a control. *Modelled*: model output driven by observations for a present or past period. *Projection*: model output for a future date or under an assumed adoption rate. *Policy-fact*: a rule, date, budget, target or institutional fact stated by the responsible authority. *Case-report*: an account of a single episode or programme by its own actors or by the press, without controls or a published protocol. *Method-reference*: a source used for its method, framework or standard, not for a result. A figure takes the grade of its weakest link, and this application's own arithmetic is shown and labelled *derived* or *indicative*.

One status sits outside those seven because it is not a grade of evidence at all. A *design-target* is a figure that states an intention rather than an estimate. The storyboard's budget section is the principal case, because it is not a budget but a set of drafting directives: equipment 'reduced to' about [250k:69.4k](AED:USD), the total 'held near' [1.4m:389k] (AED:USD), and computing to be 'raised to a realistic figure', which is a statement that the figure is not yet a realistic one [Iyyunni, 2026]. This application does not carry those directives forward. Section 2.5 (p. 12) replaces them with costed lines built from published prices and rates, with the arithmetic shown. Design-targets remain elsewhere, and are marked wherever they appear: the four month-twelve gate conditions, the milestone dates and the model validation threshold are intentions the programme undertakes to meet, set before the corridor was instrumented, and none of them is offered as evidence that it will be met [Iyyunni, 2026].

Three tiers of explanation

Three constructs carry this application: Outdoor Tourism Capacity (OTC), the Minimum Cooling Network (MCN) and the Decision Twin. The Decision Twin, on which the month-twelve gate turns, is set out at three tiers of vocabulary, after the three tiers in which Beck, McKeown and Kucan classify words: everyday words, the general vocabulary of mature readers across domains, and the vocabulary of one discipline [Beck et al., 2002].

Table 1: The ten criteria on which a full application is evaluated, in the funder's published order and wording, against the section of this application that answers each *[Dubai RDI Programme, 2025]*.

No.	Criterion	What the funder asks	Section
1	Intellectual Merit	'The extent to which the project is based on firm scientific principles and has the potential to advance fundamental knowledge in the field of interest.'	1.1 (p. 2)
2	Technical Approach / Methodology	'The soundness of the methods, experiments, observations, data, and procedures to be used to accomplish the project objectives.'	1.3 (p. 4)
3	Relevance	'The extent to which the project will contribute to progress in one or more of the Dubai RDI Priority Areas.'	1.4 (p. 6)
4	Partnerships / Capacity Building	'The potential for the project is to establish strategic partnerships between Dubai institutions of higher education and other local and international institutions.'	2.1 (p. 8)
5	Deliverables	'Does the proposal identify specific deliverables that are achievable and impactful, with timelines and milestones for completion?'	2.2 (p. 9)
6	Innovation	'The uniqueness and creativity of the planned methodology and outcomes relative to the global state-of-the-art in the field of interest.'	1.2 (p. 3)
7	Potential for Impact	'The likelihood that the project will lead to transformative progress in the relevant priority area.'	2.6 (p. 13)
8	Management Plan	'The soundness of the plan to ensure close coordination and oversight of collaborators, adherence to scheduled milestones, closure on scientific research questions, and completion of deliverables.'	2.3 (p. 9)
9	Risk / Risk Mitigation	'The extent to which the proposers have identified potential risks and described plans for mitigating those risks.'	2.4 (p. 11)
10	Budget	'The adequacy and relevance of the requested budget for supporting the proposed research and technology development, and its adherence to the budget guidelines.'	2.5 (p. 12)

A fourth line states it mathematically, so that a reviewer who wants the model does not have to infer it from the prose, and a reviewer who wants the idea does not have to decode the model. OTC and MCN are defined where they are first used, each with the unit it is counted in and, for MCN, the optimisation that defines it.

Conventions

The text is in British English: 'Dr' takes no full stop, quotations take single quotation marks, and proportions are written 'per cent'. Dates are written day, month, year (15 March 2027) and ranges take an en-dash (2027–2030); Year 1 is the first grant year, 15 March 2027 to 14 March 2028. Every money figure is printed as a pair, the dirham amount with its dollar equivalent beside it, in the form [1.4m:389k](AED:USD). The dirham has been pegged to the United States dollar at AED 3.6725 since 1997 by the Central Bank of the UAE; this document converts at the rounded working rate of AED 3.6 to the dollar so that a reader can check any conversion without a calculator, and nothing in the budget or the argument turns on the rounding. Nothing is adjusted for inflation unless the adjustment is shown. Temperatures are in Celsius (°C) except where a United States source is quoted as printed (°F). Chapter headings show the Eastern Arabic numeral beside the Western numeral, for example ٢ {2}, without the word 'Chapter'; cross-references use the Western numeral. Named metrics are set in small capitals (HLAH, Heat-Lost Activity Hours; RAH, Restored Activity Hours) and data fields in monospace (`place_id`), with the identifier schema in Appendix B (p. 24). Acronyms are expanded at first use in each chapter. One collision is unavoidable and is flagged wherever it arises: OTC means Outdoor Tourism Capacity here, but Outdoor Thermal Comfort in the Singapore urban climate twin literature [Aydt et al., 2026, Cooling Singapore 2.0, 2024].

Scope and exclusions

This application claims no new field measurement. Every quantity in it comes from the published record, from the partner agencies' own documents or from its own stated arithmetic; the first measurements of the corridor are a deliverable of Year 1 and not an input to this document. It gives no legal advice: the force-majeure and data-governance provisions referred to in Section 2.4 (p. 11) are drafting guidance subject to counsel's review. Partner agencies are described from their public documents only, and no statement here should be read as a commitment made on their behalf; the written confirmations the programme depends on are listed as conditions, not as achievements. Two extensions discussed during the design of this programme are out of scope and are not costed, staffed or claimed: the indoor work on building climate and infection intelligence, whose objective would be minimum-energy infection-resilient building operation, and the space and extreme-environment extension. Both are named once, here, so that a reviewer does not mistake their absence for an oversight.

Standing and revision

This is Version 1.0, dated 28 September 2026, and it should be cited by version and date. It states what the evidence supported on that day. Three events would justify a revision rather than an addendum: a change in the call's published rules, which are issued cycle

by cycle (grade: policy-fact) *[Dubai Future Foundation, 2025]*[2]; the signature or refusal of the four partner letters, on which the deliverables in Section 2.2 (p. 9) depend; and the outcome of the month-twelve gate in March 2028, which decides whether Phase 2 is funded at all and which this document treats throughout as a genuine stop-or-go and not as a formality.

Assumptions and Limitations

An application that declared no assumptions would be either trivial or misleading. This one carries eight, and one condition that is not an assumption at all. Each is set out below with the basis it rests on, what follows if it is wrong, and the section that would have to change. The first cycle of this initiative funded 24 of 374 proposals, about one in sixteen (derived) *[Dubai Future Foundation, 2025]*[2]; at that ratio a reviewer is entitled to have the weak joints named by the applicant rather than found by the panel.

Three statuses are kept apart throughout. An *assumption* is a claim the work carries without evidence and can retire by an act: a signature, a quotation, a supplier warranty, a season of measurement. A *dependency* is a condition outside the team's control, which no amount of work inside the project can close and which is therefore managed rather than assumed. A *limitation* is something this application does not claim, and the closing section lists those plainly, because the value of a grading scheme lies in what it refuses to grade.

One condition is carried as a dependency. The archived programme rules give grants of [500k:139k](AED:USD) to [1.5m:417k](AED:USD), an annual budget per project not exceeding [500k:139k](AED:USD), and durations of one to three years (policy-fact) *[Dubai RDI Programme, 2025]*[1]. The ask of [1.4m:389k](AED:USD) over 36 months averages [467k:130k](AED:USD) a year (derived: the total divided by three), inside that cap. What no archived document describes is a cycle receiving a full application in 2026, and the storyboard's start of 15 March 2027 presupposes one [Iyyunni, 2026]. That calendar is therefore held as an external dependency: every milestone in this application is expressed relative to the award date rather than to a fixed month, and the June to September window that the first gate condition requires is protected first, because a start that slips past the spring moves the first summer by a year and not by a quarter.

Table 2 (p. xv) is the register.

Table 2: The eight assumptions this application carries, the basis of each, what follows if it is wrong, and the section that would change. The funding calendar is not in the table: it is outside the team's control and is carried as a dependency. Section names are the evaluation criteria under which each matter is argued.

Assumption	Basis	Effect if wrong	Section affected
A working rate of AED 3.6 to the dollar	A rounded convention adopted so that any conversion can be checked without a calculator; the dirham is pegged by the Central Bank of the UAE at AED 3.6725 (policy-fact)	Every dollar figure printed here falls by about two per cent at the peg (derived, 3.6725/3.6); no dirham figure and no conclusion moves	Budget

Table 2 (continued)

Assumption	Basis	Effect if wrong	Section affected
2026 prices, with no adjustment for inflation unless the adjustment is shown	Each figure carries its price year; the dirham figure is primary throughout	Construction-linked Year 3 lines are understated if local escalation continues; equipment and personnel are unaffected inside the period	Budget
Site access from Dubai Culture and Dubai Municipality is granted by December 2026	The storyboard calendar [Iyyunni, 2026]; the letters are sought, none is held	No instrument may lawfully be placed; the first summer is lost, and a lost summer is a lost year	Management plan; Risk
The two data letters, one dataset each from the Roads and Transport Authority and the Dubai Electricity and Water Authority, conclude by February 2027	The same calendar [Iyyunni, 2026]; station exit counts with bus and abra ridership, and one summer of hourly Bur Dubai district cooling load	The model loses its exogenous demand and load inputs, the MCN optimiser its tariffs, and the fourth gate condition its counterparty	Partnerships; Risk
The validation threshold ε_{val}, set at 1.0°C on bias-corrected hourly air temperature (design-target), is reachable on this corridor	A model calibrated on one local climate zone predicted another of the same class at RMSE 0.24–0.68°C, and 0.27–1.62°C after bias correction against a 26-station network (measured) [Manandhar et al., 2019]; a Gulf heritage lane is not Abu Dhabi island	The threshold is set where the programme can pass rather than where a decision needs it, or the second gate condition fails on instrumentation rather than on science	Technical approach; Deliverables
Nine nodes, each carrying the full instrument set with black globe and natural wet bulb, fit the equipment line at [196,000.00:54,444.44] (AED:USD)	Both instruments are required by the comfort indices [Jendritzky et al., 2012, ISO, 2017b] and the storyboard's node list omits both [Iyyunni, 2026]; the twenty-node target is descoped to nine for that reason and the line built up from the unit rates registered in the companion study (derived) [Shastry et al., 2026]; no vendor quotation is held	The count gives before the instrument set does, a node that cannot compute the index not being worth its mount: the two reference-grade nodes and the paired case and control nodes are kept and an unpaired node is dropped first. Below the paired minimum the Year 3 comparison is lost rather than weakened	Budget; Technical approach
One summer of observation calibrates the model before the gate	June to September 2027 in the storyboard [Iyyunni, 2026]; the UAE precedent used four seasonal windows across two years [Manandhar et al., 2019]	An anomalous season is read as model error, or model error as weather; the Year 2 summer becomes a re-validation rather than an extension	Technical approach; Risk

Table 2 (continued)

Assumption	Basis	Effect if wrong	Section affected
A canopy over the most exposed 200 m is buildable in Year 3 under a heritage authority's consent	The storyboard's Year 3 plan [Iyyunni, 2026]; consent is sought in the December 2026 letter; no design, tender or construction budget exists	The before-and-after test against the twin's own prior prediction is lost, and Year 3 delivers a ranked candidate set instead of a measured effect	Deliverables; Risk

What is not claimed

No effect size is claimed for the intervention. This application forecasts no number of degrees and no number of restored hours for the Year 3 canopy, because the size of that effect is the quantity the experiment exists to measure and the twin's prior prediction of it is the object of the test rather than an input to it. Zone 1, the heritage core at Al Fahidi, stays untreated across all three summers as the control [Iyyunni, 2026].

The simulation in Appendix D (p. 42) does not qualify that sentence, and the distinction is worth meeting here rather than three hundred pages later. That appendix reports a modelled canopy benefit on a different corridor, in Dubai Academic City some 20 km inland, on open campus ground at a baseline sky view factor of 0.98, under a canopy geometry that is not the one Year 3 will build. It is offered as evidence that the model chain exists and runs, which is a claim about readiness and not about effect, and it is graded modelled and uncalibrated at the site because no instrument reading was taken on that corridor either. A figure carried from it to Al Ghubaiba would be precisely the transfer this register exists to refuse, and Section D.8 (p. 47) refuses it in terms.

The nearest controlled design in the evidence base published a design and not results. The Paris OASIS schoolyard programme set out its instrument specification, its paired case and control stations and its traverse protocol, and reported no effect sizes [Urban Innovative Actions, 2023, OASIS research team, 2024a, OASIS research team, 2024b]. It is used here for the design, and nowhere for a result.

The one measured-with-controls result available transfers less than it appears to. In Phoenix, the surface of coated residential streets was about 12.0°F cooler at noon and 10.5°F cooler in the afternoon than adjacent aged asphalt and 2.4°F cooler at sunrise, while air temperature at 2 m was only about 0.5°F lower in the evening and road-centre mean radiant temperature was about 5.5°F *higher* at noon and in the afternoon (measured-with-controls) [ASU and City of Phoenix, 2021]; the second phase repeated that radiant finding at +5.8°F at noon and +4.5°F in the afternoon, recorded that a road-centre reading is 'not representative of a pedestrian on an adjacent sidewalk', and put the extra heat loss a standing person must achieve at 12–13 W m^{-2} [ASU and City of Phoenix, 2024]. That is a different climate, a different surface and the road centre rather than the footway. Three consequences are carried through this application rather than set aside: shade ranks ahead of coatings in the candidate set, a coating is conditional and never preferred on air temperature alone, and the MCN optimiser carries mean radiant temperature. Neither report states a service life, a re-application interval or a unit cost, so no whole-life figure for a coating is offered here.

The megaproject overrun literature does not transfer to a grant of this size. Reference

class forecasting is used as a method [Flyvbjerg, 2014, Flyvbjerg et al., 2016, Cantarelli et al., 2025], and the distributions in it describe projects two to four orders of magnitude larger than this one. No uplift drawn from them is applied to any line of this budget; the costing is built element by element in the manner the cost-estimating guide and the life-cycle costing standard prescribe [GAO, 2020, ISO, 2017a].

No source prices a visitor-hour. OTC is therefore reported in visitor-hours and HLAH in hours lost, and neither is converted into dirhams anywhere in this application. The heat-economics literature values lost output at city scale [Arsht-Rock and Vivid Economics, 2022], the labour literature values working hours against wet-bulb globe thresholds [ILO, 2019], and the statutory midday break removes 12:30 to 15:00 from 15 June to 15 September, 2.5 hours a day across 93 days, or about 232.5 mandated no-sun hours per outdoor worker per summer (derived, indicative, on the policy-fact of the window) *[The National, 2026]*[6]. None of the three yields a price for one comfortable hour on one corridor. Where a monetary claim would be expected, the physical quantity is given and the conversion is stated to be unestablished.

Retiring an assumption

An assumption is retired by an act and not by an argument, and the acts are cheap against what they settle. A consent is retired by a signature on a letter that already has a date beside it in the calendar; a unit cost by three quotations against a written node specification; a service life by a supplier warranty; a transfer assumption by instrumenting a paired control and measuring. This register should be shorter at the month-twelve gate than it is at award, and how many rows the first year deletes is a fair measure of the first year.

[6]The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

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Acronyms

ADP approximate dynamic programming.

AI artificial intelligence.

BITS Birla Institute of Technology and Science.

CFD computational fluid dynamics.

Co-I Co-Investigator.

DEWA Dubai Electricity and Water Authority.

DFF Dubai Future Foundation.

DUCT Digital Urban Climate Twin (Cooling Singapore).

EO Earth observation.

ERA5 fifth-generation ECMWF atmospheric reanalysis.

GCC Gulf Cooperation Council.

GDP gross domestic product.

HCI Holiday Climate Index.

HLAH Heat-Lost Activity Hours.

ILO International Labour Organization.

IOC Initial Operational Capability (spiral model anchor point).

IoT Internet of Things.

IPCC Intergovernmental Panel on Climate Change.

ISO International Organization for Standardization.

LCA Life-Cycle Architecture (spiral model anchor point).

LCC life-cycle cost.

LCO Life-Cycle Objectives (spiral model anchor point).

LCZ local climate zone.

MCN Minimum Cooling Network.

MRT mean radiant temperature.

NCM National Center of Meteorology (UAE).

OGC Open Geospatial Consortium.

OSM OpenStreetMap.

OTC Outdoor Tourism Capacity (Outdoor Thermal Comfort in the Singapore urban climate twin literature).

PET physiological equivalent temperature.

PI Principal Investigator.

RAH Restored Activity Hours.

RDI Research, Development and Innovation (Dubai Future Foundation grant programme).

RMSE root-mean-square error.

RTA Roads and Transport Authority (Dubai).

SDP stochastic dynamic programming.

SVF sky view factor.

UAE United Arab Emirates.

UTC Coordinated Universal Time.

UTCI Universal Thermal Climate Index.

VoI value of information.

WBGT wet-bulb globe temperature.

List of Symbols

Costs and Economics

$C(x), w$ Life-cycle cost vector of x over capital, energy, water and maintenance; the weights applied to its components

c_{RAH} Cost per restored activity hour, the single reporting unit of the appraisal

Decisions and Policies

g_j Gate condition j at the month-twelve life-cycle architecture anchor

$x, \mathcal{X}$ A candidate set of interventions over (intervention, location) pairs; the feasible set of such sets

$x^{\dagger}$ The single intervention built and measured in Year 3

Other Symbols

$\varepsilon_{\mathrm{val}}$ Validation threshold on street-model root-mean-square error

$L(r)$ Lineage closure of a record: the transitive closure of the derivation relation over its ancestors

R_p^{*} Restored-activity-hours requirement at priority place p

State and Sets

$\mathcal{D}_t$ Observations available to the Decision Twin at epoch t

$\mathbb{E}[\cdot]$ Expectation operator

$\mathbb{N}$ Set of natural numbers

$\mathbb{R}$ Set of real numbers

p Index over places: route segments and public spaces inside the corridor boundary

Main Matter

The application itself, in ten pages and ten sections: one section for each of the ten criteria on which a full application is evaluated. The first chapter carries the research (intellectual merit, innovation against the global state of the art, technical approach and methodology, and relevance to the Cognitive Cities priority areas); the second carries the programme (partnerships and capacity building, deliverables with timelines and milestones, the management plan, risk and its mitigation, the budget, and the potential for impact).

١ {1}

The Research

Abstract

Heat removes hours from Dubai's outdoor economy, and no published system says which hours, in which places, for whom, or what returns them. Two open questions are posed here. The first is how a physically meaningful, temporally continuous climate field at 25–30 m can be inferred from observations whose native resolutions differ by orders of magnitude. The second is whether an urban area can be organised into climate-resilient cells joined by low-exposure mobility corridors. The methodology is a paired-control observation campaign on about one kilometre of Dubai Creek waterfront, a two-scale model chain validated against it to a stated numerical threshold, and a multi-objective optimiser whose output is one intervention, built and measured against the prediction made before construction.

The economy of Dubai depends on people walking, waiting, visiting and working outdoors for more of the year than heat currently allows [Iyyunni, 2026]. Climate data already exists, at the National Center of Meteorology, at the electricity and water authority and at the municipality, but it sits in separate systems and describes heat rather than what heat prevents. The distance between those two things is the research problem. The four criteria of the call that concern the science are answered here; the six that concern delivery are answered in Chapter 2 (p. 8).

1.1 Intellectual merit

The criterion is intellectual merit: 'The extent to which the project is based on firm scientific principles and has the potential to advance fundamental knowledge in the field of interest' *[Dubai RDI Programme, 2025]*[1].

The unanswered question has three parts, and none of them has a published answer for any Gulf city. Which specific activities, routes and places are lost to heat, and for how many hours a year. What economic and social value attaches to restoring them. And which single intervention restores the most activity per dirham, per litre of water and per kilowatt-hour [Iyyunni, 2026].

Every element of an answer exists somewhere in the published record; the intersection does not. Local climate zone classification coupled to microclimate simulation has

[1] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

been demonstrated in the UAE: an Abu Dhabi study joined eight Landsat-8 bands at 30 m to a 26-station urban network and to ENVI-met at 3 m per pixel (grade: measured) [Manandhar et al., 2019, Stewart and Oke, 2012]. Urban climate twins exist, Cooling Singapore's being the mature instance [Aydt et al., 2026, Cooling Singapore 2.0, 2024]; controlled before-and-after designs exist, in the Paris OASIS schoolyards [Urban Innovative Actions, 2023, OASIS research team, 2024a]; interventions measured against simultaneous untreated controls exist, in the Phoenix cool pavement trials [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]; and tourism climate indices exist at daily, destination scale [Scott et al., 2016]. No published system joins a validated street-scale comfort field to a metric of the activity heat removes, uses that metric to choose among interventions, builds one, and measures the result against the prediction made before construction. That closed loop is the contribution.

Three settled principles carry it. The human energy balance: what a standing person exchanges with a clear-sky desert street is dominated by radiation rather than air temperature, which is why both the Universal Thermal Climate Index and the wet-bulb globe temperature of ISO 7243 carry a radiant term, and why air temperature alone cannot rank an intervention [Jendritzky et al., 2012, ISO, 2017b]. Sequential decision analytics: state variable, decision, exogenous information, transition and an objective over policies, with any forecast a policy uses carried in the state [Bellman, 1957, Powell, 2019]. And the value of information: an observation earns its cost in proportion to how far it can change a decision, which makes a twelve-month campaign an instrument rather than a preliminary [Howard, 1966].

Two research questions follow, both open. The *downscaling question* asks how a physically meaningful, temporally continuous climate field at 25–30 m can be inferred from heterogeneous observations whose native resolutions differ substantially. It is not rhetorical: Landsat-8 gives 30 m multispectral bands but 100 m native thermal infrared, so a stated 30 m temperature map is not an independent thermometer in every cell (grade: method-reference) [Manandhar et al., 2019]. The credible product is a fused field at about 25–30 m, constrained by physics and downscaled by machine learning, defensible by what it is validated against rather than by the metres claimed. The *urban-cell question* asks whether an urban area can be organised into interconnected climate-resilient cells joined by optimised low-exposure mobility corridors, so that essential activity stays accessible inside a low-exposure network while accessibility, tourism and productivity are preserved. The testbed answers it for one corridor of about 800 m; no claim of generality is made in advance.

1.2 Innovation against the global state of the art

The criterion is innovation: 'The uniqueness and creativity of the planned methodology and outcomes relative to the global state-of-the-art in the field of interest' *[Dubai RDI Programme, 2025]*[1]. Table C.1 (p. 33) sets the project against the six efforts that bound it, row by row.

The scale argument is arithmetic. Destination Earth runs hourly at 5–10 km, and its extremes twin reaches 4.4 km globally and 500–750 m regionally (grade: policy-fact) *[ECMWF, 2024]*[2]. The intervention corridor is about 800 m end to end [Iyyunni, 2026], so

[2] European Centre for Medium-Range Weather Forecasts, *Fact sheet: ECMWF and the EU's Destination Earth*

at the finest regional setting the whole corridor is one or two cells. Between where those twins stop and where a shade canopy is decided lie two to three orders of magnitude, and nothing operational occupies them. Cooling Singapore reaches street scale but has not built an intervention and measured it against a prior twin prediction *[Centre for Liveable Cities, 2024]*[3]; the Paris schoolyards built and measured but registered no prediction to test [Urban Innovative Actions, 2023]; the Phoenix trials measured against simultaneous controls but ran no twin [ASU and City of Phoenix, 2021].

Two further things are new rather than merely finer. The unit of account: Cooling Singapore, Destination Earth and the Dubai platform all report environmental state, whereas this project reports the activity that state removes and returns, in Heat-Lost Activity Hours (HLAH) and Restored Activity Hours (RAH), denominated per dirham, per litre and per kilowatt-hour [Iyyunni, 2026]. And falsifiability: the Year-3 intervention is selected by the optimiser, predicted by the twin before it exists, then measured against an untreated control across one summer, so the twin can be found wrong on the record.

1.3 Technical approach and methodology

The criterion is technical approach and methodology: 'The soundness of the methods, experiments, observations, data, and procedures to be used to accomplish the project objectives' *[Dubai RDI Programme, 2025]*[1].

Understanding: Decision Twin

TIER 1: IN PLAIN WORDS

A place to try an intervention before it is built. Describe a canopy, a row of trees or a change of surface, and it returns how many comfortable hours that would give back, at what cost in money, water and power, and how sure it is. It is not a three-dimensional picture of the city.

TIER 2: IN PROFESSIONAL TERMS

A simulation and decision environment that compares candidate interventions against a common baseline and reports the comparison as a short brief for the person deciding, distinct from a rendering and distinct from the geometry platform it sits on [Iyyunni, 2026]*[Dubai Media Office, 2026]*.

TIER 3: IN TECHNICAL TERMS

A predictive layer over a validated microclimate field returning, for each candidate intervention set, a distribution over restored activity hours and over the resources consumed, with the provenance of every input carried per record, so that a brief can state what is observed, what is modelled and what is assumed [Powell, 2019, W3C, 2013].

AS A MATHEMATICAL MODEL

For a candidate $x \in \mathcal{X}$ and observations $\mathcal{D}_t$ the twin returns $\widehat{\Delta\mathrm{RAH}}(x) = \mathbb{E}\big[\Delta\mathrm{RAH}(x) \mid \mathcal{D}_t\big]$ with its predictive interval. The Year-3 test is whether the

initiative, ECMWF media centre, 2024. https://www.ecmwf.int/en/about/media-centre/focus/2024/fact-sheet-ecmwf-and-eus-destination-earth-initiative

[3] Centre for Liveable Cities, *Box Story: Cooling Singapore*, Knowledge Hub, Singapore, 2024. https://knowledgehub.clc.gov.sg/liveability-in-action/box-story--cooling-singapore/

measured $\Delta\mathrm{RAH}^{\mathrm{obs}}(x^{\dagger})$ for the one built intervention $x^{\dagger}$ falls inside the interval recorded before construction.

Figure C.1 (p. 34) draws the layers and marks six things routinely treated as one. Each is a separate deliverable with its own owner and its own way of failing [Iyyunni, 2026]: a data model fails silently when a field changes meaning, a semantic model when two agencies mean different things by one term, a knowledge graph on unprovenanced links, an AI model by being confidently wrong, a twin when it is used unvalidated, and a decision system when nobody acts on it.

The observation design. Nine internet-of-things (IoT) microclimate nodes and one reference station sample air temperature, humidity, wind, radiation and surface temperature every ten minutes across both zones. The storyboard proposed twenty [Iyyunni, 2026]; nine that carry the full instrument set are worth more than twenty that cannot compute a heat-stress index, and Section 2.5 (p. 12) takes that descope in the open. Zone 1, the heritage core at Al Fahidi, has narrow lanes, coral-stone walls and wind towers, and is already passively cooled: it is the calibration baseline, takes no intervention in any of the three years, and is the untreated control. Zone 2, about 800 m of open road and pavement from Al Ghubaiba metro station to that core, is the intervention site. Every intervention segment carries a paired case and control node rather than relying on the single reference station, so that the Year-3 effect is the change at the treated point net of the simultaneous change at its control [OASIS research team, 2024a].

The instrument correction. One change to the node specification is a condition of the science. As drafted it lists neither a black-globe nor a natural-wet-bulb thermometer [Iyyunni, 2026]; without them wet-bulb globe temperature cannot be computed to ISO 7243 and mean radiant temperature is not measured at all [ISO, 2017b]. Both are added to the reference station and to every paired case and control node, at [4.5k: 1.25k](AED:USD) an assembly and [31.5k:8.75k](AED:USD) across the network (derived). Wet-bulb globe temperature is then reported beside the Universal Thermal Climate Index, because the labour series ministries read are stated in the former while the pedestrian index is the latter [ILO, 2019, Jendritzky et al., 2012].

The model chain and its validation target. A district model runs at 25–50 m and a street model at 1–5 m, with machine-learning downscaling and empirical bias correction between and beneath them [Iyyunni, 2026]. The validation target is a number, not an intention: hourly bias-corrected air temperature from the street model must reach a root-mean-square error $\varepsilon_{\mathrm{val}}$ of 1.0 °C or better against the node network across the summer of 2027, reported separately for surface temperature. The threshold comes from the one UAE precedent, the Abu Dhabi coupling's 0.27–1.62 °C after bias correction across twenty district-seasons (grade: measured) [Manandhar et al., 2019]: it sits inside a demonstrated band and is neither a formality nor a guess. It is the second of the four month-twelve gate conditions [Iyyunni, 2026].

The optimiser. The candidate set is shade canopy, tree planting, cool pavement, misting rest point and revised route. The optimiser searches binary decisions x over (intervention, location) pairs against a life-cycle cost vector $C(x)$ carrying capital, energy, water and maintenance in the sense of ISO 15686-5 [ISO, 2017a], and returns the Minimum Cooling Network (MCN), $x^* = \arg\min_x w^\top C(x)$ subject to $\Delta\mathrm{RAH}_p(x) \geq R_p^*$ at every priority place p: the smallest sufficient network rather than the largest affordable one.

Why mean radiant temperature decides. The Phoenix trials settle the ranking quantity. Coated residential streets ran about 12.0 °F cooler at noon and 10.5 °F cooler in the afternoon than adjacent aged asphalt, and 2 m air temperature about 0.5 °F lower in the evening; road-centre mean radiant temperature was about 5.5 °F *higher* at noon and in the afternoon (grade: measured-with-controls) [ASU and City of Phoenix, 2021]. Phase II reproduced it at +5.8 °F at noon and +4.5 °F in the afternoon, and put the pedestrian cost at 12–13 W m^{-2} of additional heat loss demanded of a standing person (grade: measured-with-controls) [ASU and City of Phoenix, 2024]. An optimiser scoring on air temperature would call that coating a success while making the walk worse. Shade therefore ranks first and coatings enter conditionally; neither report states a service life, a re-application interval or a unit cost, so no whole-life figure for coatings is asserted.

The chain is built, and has been run. None of the above is a promise of software. The model chain described here has been implemented and run end to end on a second Dubai corridor, 495 m of walking route in Dubai Academic City between The Myriad and the BITS Pilani Dubai Campus, and Appendix D (p. 42) is the record of it: eleven pipeline steps from source download to a diurnal comfort field, conforming to the geodetic and interchange standards rather than intending to, with the sensitivity of the answer to each assumption measured rather than asserted and a published canopy specification tested against the chain and found wanting on its own terms. Two consequences bear on this criterion. The twelve-month gate is not being asked to deliver a first implementation under field conditions, which is where the schedule risk in Section 2.4 (p. 11) would otherwise sit. And the grant buys instruments, access and validation rather than a model chain, which is the half that is missing: every atmospheric quantity in that dry run is reanalysis from a grid cell 1.81 km away, and not one instrument reading was taken at the corridor. Appendix D (p. 42) carries no prediction for the intervention this application will build, and Section D.8 (p. 47) states why it cannot.

1.4 Relevance to the Cognitive Cities priority areas

The criterion is relevance: 'The extent to which the project will contribute to progress in one or more of the Dubai RDI Priority Areas' *[Dubai RDI Programme, 2025]*[1]. The priority area is Cognitive Cities, whose four topics are 'Smart mobility'; 'Next-generation built environments and infrastructure'; 'Climate-neutral and resilient urban systems'; and 'Urban health intelligence for proactive well-being' *[Dubai RDI Programme, 2025]*[1].

The first topic is served at the point where it is usually lost: the corridor is a first-and-last-mile link between a metro station and a destination, and station exit counts with bus and abra ridership are the demand series against which comfort is read [Iyyunni, 2026]. At scale that demand is the Dubai Walk master plan, 3,300 km of new and 2,300 km of

renovated walkways by 2040 (grade: policy-fact) *[Dezeen, 2024]*[4]*[Khaleej Times, 2025]*[5]; a walkway nobody can use in August is a capital asset with a seasonal outage, which is the second topic: shade, surface and planting decisions tested before capital is committed. The third is served by the denominators, since RAH per kilowatt-hour and per litre is what stops comfort being bought with cooling load, and the hourly Bur Dubai district cooling load supplies the energy denominator [Iyyunni, 2026]. The fourth is served as an exposure layer, not a health product: work is banned under direct sunlight and in open spaces from 12:30 to 15:00 between 15 June and 15 September (grade: policy-fact) *[The National, 2026]*[6], about 232.5 mandated no-sun hours per outdoor worker per summer (derived and indicative: 2.5 hours across 93 calendar days). The indoor extension of this architecture, Minimum-Energy Infection-Resilient Building Operation, is a named non-goal here.

Of the cross-cutting technologies the call names, the work sits under 'Internet of Things (IoT), edge computing, and digital twins' and draws on 'Artificial intelligence' for the downscaling and the optimisation *[Dubai RDI Programme, 2025]*[1].

The strongest relevance argument is the platform that already exists. The Dubai Digital Twin Platform launched on 2 July 2026 with three-dimensional models of more than 195,000 buildings and more than 280,000 infrastructure assets across more than 1,500 geospatial layers, and rainfall as its named physics-based simulation (grade: policy-fact) *[Dubai Media Office, 2026]*[7]. This project does not build a competing twin; it builds the decision layer for geometry the emirate has already paid for, the hour-by-hour comfort field that platform does not carry and the brief that turns it into a choice. Anchoring there converts a platform-build risk into a data request, and it is why the work fits inside [1.4m: 389k](AED:USD) over 36 months rather than an order of magnitude more.

[4]Dezeen, *Dubai Walk masterplan to make city pedestrian friendly*, 17 December 2024. https://www.dezeen.com/2024/12/17/dubai-walk-masterplan-pedestrian/

[5]Khaleej Times, *Walk in the sky? How Dubai's Future Loop will change commuting*, 2025. https://www.khaleejtimes.com/uae/transport/dubai-future-loop-project-connect-10-landmarks-ease-traffic

[6]The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

[7]Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

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The Programme

Abstract

This chapter answers six of the ten criteria on which a full application is scored: partnerships and capacity building, deliverables, management, risk, budget and potential for impact. It names the lead institution, the collaborating entity carrying its own budget line, and the four agencies with the single request made of each. It sets the phase structure against fourteen dated milestones, each with the test by which completion is judged rather than asserted. It maps the spiral model's anchor points onto three yearly loops, with the month-twelve gate as the life-cycle architecture anchor. It registers ten risks with named owners. And it sets a budget of [1.4m:389k] (AED:USD) inside the annual cap, 74.3 per cent of it utilised within Dubai.

The programme is bounded on purpose: one corridor, nine instrumented nodes, four agencies with one request each, two funded posts, one built structure, one measured comparison. The funder's first cycle selected 24 proposals from 374 (policy-fact), about one in sixteen (derived) *[Dubai Future Foundation, 2025]*[1]. At that rate every commitment below carries the test that would settle it and the name of whoever owns it.

2.1 Partnerships and capacity building

The criterion is 'The potential for the project is to establish strategic partnerships between Dubai institutions of higher education and other local and international institutions'*[Dubai RDI Programme, 2025]*[2].

The lead institution is BITS Pilani Dubai Campus, where the Principal Investigator and four Co-Investigators are fully affiliated and carry mobility, the built environment, edge sensing and systems integration. The collaborating entity is Curtin University Dubai, contributing Green AI, machine-learning downscaling and the multi-objective optimiser and holding its own budget line of [140k:38.9k](AED:USD): a letter of support carrying no money is not a partnership.

[1] Dubai Future Foundation, *Dubai's RDI Grant Initiative Returns to Fund Groundbreaking Research*, news, June 2025. https://www.dubaifuture.ae/latest-news/dubais-rdi-grant-initiative-returns-to-fund-groundbreaking-research/

[2] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

Each agency is asked for one page-sized request, because one that would need an internal data-sharing project of its own does not get signed. The Roads and Transport Authority: Al Ghubaiba station exit counts with bus and abra ridership. The Dubai Electricity and Water Authority: the Bur Dubai district cooling load, hourly, for one summer. Dubai Culture: permission to instrument the heritage core at Al Fahidi. Dubai Municipality: one letter covering permission to instrument the corridor and access to the geospatial layers of the Dubai Digital Twin Platform launched on 2 July 2026 (policy-fact) *[Dubai Media Office, 2026]*[3]. Consent to build is separate, and is sought on selection in November 2028.

All six institutional support measures the call requires are answered, and Section C.3 (p. 32) sets out how *[Dubai RDI Programme, 2025]*[2]. Two bind hardest. Fast-tracked procurement: the sensing line must reach purchase order within six weeks of grant start, or the first summer is lost. Reduced teaching load: the contact-hour reduction for five people over 36 months is [432k:120k](AED:USD) to [864k:240k](AED:USD) of institutional time, not claimed from the grant (derived) [Shastry et al., 2026].

The capacity built is two people: a doctoral student in exposure and behaviour analysis for all 36 months, and a postdoctoral researcher in climate modelling and validation for 34, from month three. Both are resident in Dubai and together cost [900k:250k](AED:USD), 64.3 per cent of the ask (derived). A grant of this size produces not a platform but two trained people and a dataset.

No letter is held today, and the application says so rather than implying otherwise. The two institutional letters are signed before submission; the Dubai Culture and Dubai Municipality letters are due in December 2026, after a corridor walk with all four agencies; the two data letters and the Curtin University Dubai agreement, carrying its signed budget share, are due in February 2027, before the grant begins.

2.2 Deliverables, timelines and milestones

The criterion is 'Does the proposal identify specific deliverables that are achievable and impactful, with timelines and milestones for completion?'*[Dubai RDI Programme, 2025]*[2].

Fourteen dated milestones run from the pre-award letters to grant close, and each states how completion is tested, because a milestone whose acceptance test is its own assertion cannot release money. Table C.2 (p. 36) in Appendix C (p. 32) gives all fourteen with their tests; Table 2.1 (p. 10) gives the shape.

2.3 Management plan

The criterion is 'The soundness of the plan to ensure close coordination and oversight of collaborators, adherence to scheduled milestones, closure on scientific research questions, and completion of deliverables'*[Dubai RDI Programme, 2025]*[2].

The programme runs as three yearly spiral loops, each closing on an anchor point [Boehm and Hansen, 2000]. The pre-award loop closes at *life-cycle objectives* on 15 March 2027: one feasible architecture, named partners, one written win condition per partner.

[3] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

Table 2.1: The phase structure, and the deliverable whose acceptance test closes each phase. The fourteen milestones behind this summary, each with its acceptance test, are in Table C.2 (p. 36). Dates and phases are the storyboard's and are design-targets [Iyyunni, 2026]; the acceptance tests are the applicants' own.

Phase	Period	Closing deliverable	Test that closes it
Pre-award	to Mar 2027	Site-access, data and collaboration letters	Each names its dataset or its marked node positions, its period and a contact; the collaboration agreement carries a signed, non-zero budget share
Phase 1	Mar 2027 – Mar 2028	Hour-by-hour comfort map of both zones, validated	Bias-corrected hourly air temperature within 1.0°C root-mean-square error on a held-out week [Manandhar et al., 2019]
Gate	Mar 2028	Stop-or-go decision at month twelve	All four gate conditions evidenced in one pack; any one failing closes the project with the dataset and platform as deliverables
Phase 2	Mar 2028 – Mar 2029	MCN for the corridor; Decision Twin version 1; the intervention selected	A named officer at a partner runs a scenario comparison unaided and receives a brief of two pages or fewer; the sustainment commitment is attached to the selection
Phase 3	Mar 2029 – Mar 2030	One intervention built; before-and-after comparison; decision briefs and method document	Difference-in-differences against the untreated control, reported against the prediction lodged before construction whether or not the two agree

Loop 1 closes at *life-cycle architecture* in March 2028, the month-twelve gate, which is where resource commitment escalates and the major risks must therefore be resolved rather than listed. Loop 2 closes in March 2029 with the intervention selected, its written sustainment commitment attached and its consent to build in hand. Loop 3 closes at *initial operational capability* between September 2029 and February 2030, when something exists that the public walks under and it has been measured.

The four gate conditions, and the one the team cannot satisfy alone

Phase 2 funds are released only if all four hold (design-targets, as the storyboard writes them) [Iyyunni, 2026]: g_1, the network live at 90 per cent uptime or better across the summer; g_2, the model within the stated error of the sensors; g_3, the comfort map of both zones delivered hour by hour; g_4, one of the transport or the utility authority confirming in writing that the output is useful for a decision it faces. Release requires $\bigwedge_{j=1}^{4} g_j$. The first three are engineering. The fourth is another organisation's judgement, the one condition the team cannot satisfy by working harder. A failed gate closes the project with the dataset and the platform as its deliverables: planned, not improvised.

Coordination is thin and frequent. Lead and collaborator meet monthly against one shared repository and one data dictionary, so that downscaling and sensing cannot drift into different definitions of a place or an hour. The four partners are convened quarterly, each review showing a partner the metric computed on its own data. Reporting has three cadences: uptime and model error published weekly through each summer, so that a falling trend shows in June rather than in September; a quarterly progress and expenditure report to the funder; and the gate pack six weeks before the gate.

2.4 Risk and risk mitigation

The criterion is 'The extent to which the proposers have identified potential risks and described plans for mitigating those risks'*[Dubai RDI Programme, 2025]*[2].

Ten risks are registered in the form of ISO 31000, each with a named individual rather than a committee against it [ISO, 2018]; Table C.3 (p. 38) in Appendix C (p. 32) carries the register with its mitigations and owners. Four dominate. Sensor uptime below 90 per cent across a Gulf summer is the only entry rated high on both likelihood and impact: dust, airborne salt and air temperatures above 45°C are the named failure modes, the enclosure specification is written against them, two spare nodes are held and uptime is published weekly. Scope exceeding the envelope is the second, answered by descopes taken in the open in Section 2.5 (p. 12) rather than by a reserve. Model error above the threshold is the third, answered by fixing the threshold in advance at 1.0°C and reporting the shortfall as a finding if it is missed. The fourth is the one below.

The largest exposure is a sentence somebody else has to write

Of the ten entries, g_4 is the one that closes the programme and the one the team cannot close by working harder. A written confirmation that the output is useful for a decision an authority actually faces is a judgement, not a deliverable, and it will not

be given on the strength of a map handed over in February 2028. The treatment is to make it a design input: the decision each authority faces is elicited before award and written into the win-conditions table; the comfort map is specified in that decision's own units; a draft is shown in December 2027 with two months in hand; and both authorities are asked rather than one. Stakeholders cannot endorse a useful product until they have seen one [Boehm et al., 1998].

One risk is treated by the schema rather than by a plan. Pedestrian counts arrive from the transport authority already aggregated to the place-hour; no image, device identifier or person-level key is retained; and the `is_synthetic` firewall keeps modelled records out of measured series, so a simulated count can never be reported as an observation [W3C, 2013]. Appendix B (p. 24) carries the fields.

2.5 Budget

The criterion is 'The adequacy and relevance of the requested budget for supporting the proposed research and technology development, and its adherence to the budget guidelines' *[Dubai RDI Programme, 2025]*[2].

The ask is [1.4m:389k](AED:USD) over 36 months, inside the envelope of [500k:139k](AED:USD) to [1.5m:417k](AED:USD) *[Dubai RDI Programme, 2025]*[2]. The storyboard's budget section stated drafting directives, equipment 'reduced to' about [250k:69.4k](AED:USD) and computing to be 'raised to a realistic figure' [Iyyunni, 2026]; those are design-targets, and the costed lines that replace them are built from unit prices in Section C.6 (p. 40). Table C.4 (p. 40) there gives the full breakdown by category and year; Table 2.2 (p. 12) here gives the two tests it has to pass.

Table 2.2: The budget by year, 2026 prices, against the two tests the guidelines set. Every figure is derived, on the unit rates registered in the companion study [Shastry et al., 2026] and built up line by line in Table C.4 (p. 40). An average year is not a year: the ask divided by three gives [467k:130k](AED:USD), and a profile built on that average would have breached the cap in Year 1, when the instrumentation is bought.

	Year 1	Year 2	Year 3	Total
Equipment and instrumentation	196,000	–	–	196,000
Personnel, two funded posts	276,000	312,000	312,000	900,000
Collaborator share	–	90,000	50,000	140,000
Computing, travel, governance	28,000	78,000	58,000	164,000
Annual total, AED	**500,000**	**480,000**	**420,000**	**1,400,000**
Annual cap, AED	500,000	500,000	500,000	–
Within cap	**Yes**	**Yes**	**Yes**	–
Spent within Dubai, AED	Personnel and collaborator share			**1,040,000**
Dubai share	Required at least 50 per cent			**74.3%**

Two compliance tests, both passed on the face of the table. No project year exceeds the cap of [500k:139k](AED:USD), Year 1 sitting exactly on it because that is when the

instrumentation is bought, and the three years sum to [1.4m:389k](AED:USD) (derived). At least 50 per cent must be utilised within Dubai *[Dubai RDI Programme, 2025]*[2]: the two personnel lines and the collaborator share, [1.04m:289k](AED:USD), are spent in the emirate by residents of it, which is 74.3 per cent, and procuring the instrumentation through a supplier registered in Dubai would raise it to 88.3 per cent (derived).

Four descopes make the scope fit, each a decision taken rather than an omission discovered: nine instrumented nodes rather than twenty, because nine that compute the index beat twenty that cannot; no field workstation; no contracted platform in Year 1; and the postdoctoral post beginning in month three. Two things sit outside the grant: the built structure, and all operation, water, energy and repair after March 2030, both belonging to the owning authority under the life-cycle convention this programme reports in [ISO, 2017a]. Where ambition exceeds the envelope the response is a stated descope, not a declared reserve.

2.6 Potential for impact

The criterion is ‘The likelihood that the project will lead to transformative progress in the relevant priority area’*[Dubai RDI Programme, 2025]*[2].

Four beneficiaries, each gaining a quantity it does not have today. Government gains a ranked option set in which shade, tree planting, surface material and rest points are compared by RAH, Restored Activity Hours, per dirham of life-cycle cost, before capital is committed. Tourism gains OTC in visitor-hours per place per day of a given kind, and the HLAH an intervention returns. Mobility gains the first-and-last-mile link between Al Ghubaiba station and the heritage core ranked by exposure hour by hour, against a plan for 3,300 km of new and 2,300 km of renovated walkways by 2040 (policy-fact) *[Dezeen, 2024]*[4] that will otherwise be routed without a comfort layer. Resources gains RAH per kilowatt-hour of district cooling load and per litre of water, which is what tells an option that adds comfort apart from one that moves the load indoors.

The publication window is external and fixed. The Special Report on Climate Change and Cities is scheduled for approval in March 2027, before this programme’s first summer, so nothing measured here can enter it (policy-fact) *[IPCC, 2026b]*[5]*[IPCC news, 2024]*[6]. The seventh-cycle Working Group assessment is the target instead, and the September 2029 before-and-after result with the February 2030 method document is what the programme intends to put before it (design-target).

Sustainment is answered as a condition rather than a hope. Procurement by a Dubai agency, on a second corridor, is the stated sequel to this grant [Iyyunni, 2026]. In the WinWin case study only one of five applications carried forward was implemented, the one with a budget, people and facilities to sustain it, and sustainment commitment became the leading selection criterion the year after (case-report) [Boehm et al., 1998]. The Year-2

[4] Dezeen, *Dubai Walk masterplan to make city pedestrian friendly*, 17 December 2024. https://www.dezeen.com/2024/12/17/dubai-walk-masterplan-pedestrian/

[5] Intergovernmental Panel on Climate Change, *Special Report on Climate Change and Cities*, IPCC Seventh Assessment Cycle; approval scheduled March 2027. https://www.ipcc.ch/report/special-report-on-climate-change-and-cities/

[6] Intergovernmental Panel on Climate Change, *IPCC approves outlines of the first two reports in the seventh assessment cycle*, IPCC news, 2 August 2024. https://www.ipcc.ch/2024/08/02/ipcc-approves-outlines-of-the-first-two-reports-in-the-seventh-assessment-cycle/

intervention choice is therefore made subject to a written sustainment commitment from the owning authority, attached to the selection in November 2028 and tested at that milestone. An asset nobody has agreed to operate is not an impact; it is a liability with a ribbon on it.

End Matter

The source and evidence register, against which every figure in this application can be traced back to a document of stated standing, together with the record of what could not be archived at all; the identifier schema on which life-cycle measurement depends, which should be read as a procurement document; the programme detail, which carries at full width the five exhibits the ten-page main matter cites and has no room to print; and the simulation, which is the model chain of Section 1.3 (p. 4) already built and already run end to end on a second Dubai corridor, offered as evidence that the methodology exists and never as a prediction of what the Year 3 intervention will do. Then the Glossary, the Bibliography and the Index.

A {A}

Source and Evidence Register

Abstract

Fifty-one sources carry every quantity in this application, and this register is the authoritative statement of what each of them can bear: its kind, its evidence grade, the claim it supports here and the section that uses it. Two further sources are named in the text but were not reviewed, and are registered plainly as such, together with the entries held from their publication details alone. The register then records the defects that make particular numbers unsafe to quote: the Phoenix city-wide water and energy extrapolations from a single day, the absent effect sizes at Paris OASIS, the Makkah cooling figures that rest on agency statements with no published protocol, and megaproject cost statistics two to four orders of magnitude out of scale. It closes with the evidence horizon, 22 September 2026, the date on which the companion feasibility study secured and dated every source used here.

An assertion that cannot be re-checked is not evidence, whatever its footnote says. A reviewer should be able to take any figure in this application, find the document behind it, and see what grade of evidence the applicants thought that document carried before they wrote the number down. This appendix provides those three things for every source used. It is deliberately mechanical: source, kind, grade, extent, section. Nothing in it argues for the project; everything in it is a handle by which an argument made in the preceding sections can be pulled apart.

The register is also the authoritative list. Where a section names a source, what that source can support is settled here and not there, and a figure whose source does not appear in this register has not been cleared for use.

A.1 How to read the register

The *Kind* column reads *bib.* for a bibliography entry and *web* for a web source, and the distinction is evidential. A bibliography entry has a normative identity: a journal article or book cited by its digital object identifier, a numbered standard whose designation is its citation, or a report formally issued by a named institution. It is listed at the back without a web address, because those details are enough to obtain it. A web source (a press release, a news report, a programme page) has no such identity and is carried instead as a bracketed label with a live link at its first use. Web sources establish that a programme exists or that an authority stated a rule; they are almost never evidence for an effect size, and every quantity

graded measured or measured-with-controls here rests on a bibliography entry.

The *Evidence grade* column is the only one carrying a judgement. *Measured-with-controls*: an effect measured against a simultaneous or paired untreated comparator. *Measured*: an instrumented observation without an intervention contrast. *Modelled*: simulation calibrated on observations. *Projection*: a forward scenario under assumed parameters. *Policy-fact*: a rule, date, budget or institutional fact stated by the responsible authority. *Case-report*: a single narrated episode without a counterfactual. *Method-reference*: a specification of procedure that asserts nothing empirical. A figure takes the grade of its weakest link; this application's own arithmetic is shown and labelled derived or indicative; and a figure stating an intention rather than an estimate is marked a design-target, which is not a grade of evidence at all.

The *Section* column gives the criteria on which the application is organised, in the funder's own order: 1 intellectual merit, 2 technical approach, 3 relevance, 4 partnerships and capacity building, 5 deliverables, 6 innovation, 7 potential for impact, 8 management plan, 9 risk and its mitigation, 10 budget *[Dubai RDI Programme, 2025]*[1].

Table A.1 (p. 18) registers fifty-one sources, thirty-four bibliography entries and seventeen web sources (derived, by tally over the table). Their distribution is worth reading before the table itself: twenty-two are method-references, ten are policy-facts, seven are case-reports, four are measured, three are modelled, three are projections and two are measured-with-controls (derived). Both of the last two concern cool pavement in Phoenix, and both are cited here for a result that argues against ranking coatings first. That is the evidential position of the field, and it is the standard the Year-3 before-and-after campaign is designed to meet rather than to cite.

[1] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

Table A.1: Source and evidence register. *Kind*: bib. is a bibliography entry, web a footnoted web source. *Section* numbers are the funder's ten evaluation criteria in the order this application answers them *[Dubai RDI Programme, 2025]*.

Source	Kind	Evidence grade	Extent: what it supports here	Section
The application's own foundations				
Iyyunni, 2026	bib.	policy-fact (design document)	Scope, testbed, partners, gate conditions, funded posts and budget intent	1–3, 5–10, Pref., App. B
Shastry et al., 2026	bib.	method-reference	The grading scheme used here, the source archive, the life-cycle cost and risk methods	4, 9, 10, Pref.
Urban heat: field evidence and models				
ASU and City of Phoenix, 2021	bib.	measured-with-controls	Same-day coated against aged asphalt: surface and air temperature, road-centre mean radiant temperature, reflectivity decay	1, 2, Pref., App. B
ASU and City of Phoenix, 2024	bib.	measured-with-controls	Phase II mean radiant temperature at noon and afternoon; the pedestrian caveat that ranks shade before coatings	1, 2, 5, Pref., App. B
Manandhar et al., 2019	bib.	measured	Landsat-8 local climate zones with a 26-station network coupled to simulation at 3 m; the error band the month-12 gate adopts	1, 2, 5, 9, Pref., App. B
OASIS research team, 2024 (microclimate)	bib.	method-reference	Paired case-and-control station design, traverse protocol and the comfort-capable node specification	1, 2, Pref.
OASIS research team, 2024 (paving)	bib.	measured	Thermal behaviour of paving materials in the candidate intervention set	Pref., App. A
Urban Innovative Actions, 2023	bib.	case-report	The OASIS programme, its stakeholder walk-through and its before-and-after design	1, Pref.
Cooling Singapore 2.0, 2024	bib.	modelled	The nearest comparable urban climate twin, and the OTC vocabulary collision this application must resolve	1, Pref.
NSCC Singapore, 2024	bib.	case-report	The scale of computing a twin of that kind requires	App. A
Stewart and Oke, 2012	bib.	method-reference	The local climate zone classification scheme	1, App. B
Heat, labour and the economics of lost activity				
ILO, 2019	bib.	projection	More than 2 per cent of UAE GDP lost to heat stress at work by 2030; the metabolic classes behind HLAH	1, 3, 7
Romanello et al., 2024	bib.	modelled	Hours of heat-stress risk for light outdoor activity: the visitor-side ancestor of HLAH	1, 7
Arsht-Rock and Vivid Economics, 2022	bib.	modelled	City-scale output lost to heat as context; main report not obtained (Table A.2 (p. 21))	3, 7
Vivid Economics, 2022 (methodology)	bib.	method-reference	The lost-hour threshold convention and its treatment of the hottest hours	1, 2
Scott et al., 2016	bib.	method-reference	The Holiday Climate Index: the destination-scale incumbent that OTC must improve upon	1, 6

continued on next page

Source	Kind	Evidence grade	Extent: what it supports here	Section
Indices, standards and provenance				
Jendritzky et al., 2012	bib.	method-reference	The rationale for the Universal Thermal Climate Index, adopted for pedestrians	2
ISO 7243:2017	bib.	method-reference	Wet bulb globe temperature; the instrument requirement that fixes the equipment line	2, 10
W3C, 2013	bib.	method-reference	The provenance pattern carried on every record of the data layer	2, 8
ISO 31000:2018	bib.	method-reference	Risk vocabulary and process for the risk register	9
ISO 22301:2019	bib.	method-reference	Continuity vocabulary for the loss and recovery plan	9
ICC, 2020	bib.	method-reference	Force-majeure and hardship drafting for the partner agreements	9
ISO 15686-5:2017	bib.	method-reference	Life-cycle costing vocabulary; preview only, so no service life or rate is taken from it	10
Method: decisions, development and cost				
Boehm and Hansen, 2000	bib.	method-reference	Spiral anchor points and risk-driven effort; the month-12 gate as a life-cycle commitment point	5, 8
Boehm et al., 1998	bib.	case-report	The WinWin case, in which only the application with a sustaining budget survived	4, 8
Powell, 2019	bib.	method-reference	The sequential decision framework behind the MCN optimiser	1, 2
Powell, 2022	bib.	method-reference	The same framework in book form; held as metadata only	2
Bellman, 1957	bib.	method-reference	Dynamic programming, reached through later citations	2
Howard, 1966	bib.	method-reference	Expected value of information: why the cheapest most informative step comes first	5, 8
GAO, 2020	bib.	method-reference	Cost-estimating practice, and why catastrophic events are handled contractually	9, 10
Flyvbjerg, 2014	bib.	measured	Cost and schedule overrun distributions, as context only (Section A.3 (p. 21))	9
Flyvbjerg et al., 2016	bib.	measured	Overruns on fixed-deadline programmes, as context for the summer window	9
Cantarelli et al., 2025	bib.	method-reference	Reference class forecasting, its promises and its documented problems	10
Beck et al., 2002	bib.	method-reference	The three vocabulary tiers used in the Understanding boxes	front matter
The call, the city and the region				
[Dubai RDI Programme, 2025]	web	policy-fact	Envelope, annual cap, duration, eligibility, the Dubai-spend condition, the institutional support measures, the priority topics and the ten criteria	1–10

continued on next page

Source	Kind	Evidence grade	Extent: what it supports here	Section
[Dubai Future Foundation, 2025]	web	policy-fact	The preceding cycle's calendar, and 24 awards from 374 proposals, about one in sixteen (derived)	3, 10
[Dubai Media Office, 2026]	web	policy-fact	The Dubai Digital Twin Platform launched 2 July 2026, with more than 195,000 buildings and more than 1,500 geospatial layers: the platform this twin sits on	3, 4, 7
[Dezeen, 2024]	web	policy-fact	Dubai Walk: 3,300 km of new and 2,300 km of renovated walkways by 2040, the demand at scale	3, 7
[Khaleej Times, 2025]	web	policy-fact	The pedestrian programme's stated ambition and its named links	3, 7
[URB, 2025]	web	projection	A 93 km climate-controlled corridor concept: the alternative this application is not proposing	3, 6
[The National, 2026]	web	policy-fact	Work banned under direct sunlight and in open spaces from 12:30 to 15:00, 15 June to 15 September	1, 3, 7
[MSE Singapore, 2023]	web	policy-fact	The heat stress advisory and its published weighting of natural wet-bulb, black-globe and dry-bulb temperature	2, 6
[Centre for Liveable Cities, 2024]	web	case-report	Cooling Singapore as a programme, its scope and its institutional form	6
[ECMWF, 2024]	web	method-reference	Destination Earth at 4.4 km globally and 500–750 m regionally: the scale at which continental twins stop	1, 6
[IPCC news, 2024]	web	policy-fact	The seventh assessment cycle's report outlines and their timing	7
[IPCC, 2026b]	web	policy-fact	The Special Report on Climate Change and Cities and its publication window	7
[Saudipedia, 2025]	web	projection	Irrigation network and treated-wastewater targets: a regional comparator with a water cost	3, 7
[Arab News, 2024]	web	case-report	Road cooling for pilgrims, reported without a protocol (Section A.3 (p. 21))	6
[Gulf News, 2025]	web	case-report	The stated expansion of cooled and rubberised roads	6
[Saudi Gazette, 2024]	web	case-report	Cooling measures at the holy sites, as announced	6
[Uber, 2025]	web	method-reference	Cell areas and edge lengths by resolution, for the place identifier	2

A.2 Sources not reviewed, and sources held as metadata only

Two sources named in this application could not be obtained at all, and three more are held only in part: a main report for which the public methodology stands in, a standard held as a preview, and a standard that has still to be purchased. The rule is stated once here: an unreviewed source may be named, its title given and its existence asserted, but no number, method or result may be attributed to it, and the text must say that it was not reviewed. Table A.2 (p. 21) records each with its status and with what follows for the claim it might otherwise have supported. The first row is the one a Dubai reviewer will look for: the single Dubai-specific local climate zone study in the literature, cited by title alone and yielding nothing.

Table A.2: Sources named in this application that were not reviewed or were obtained only in part, with the consequence for what may be claimed.

Source	Status	Consequence here
Bande et al., 2020, local climate zones in relation to ENVI-met in Dubai, `dubai2020lcz`	Not reviewed; publisher verification defeated retrieval	Named as the one Dubai-specific precedent; no result, parameter or error figure is taken from it, and the gate's error band is taken from the Abu Dhabi study instead
Aydt et al., 2026, the Singapore Digital Urban Climate Twin paper, `duct2026paper`	Not reviewed; publisher bot wall	The Singapore twin is described from the poster and the computing note, which are held
Arsht-Rock, *Hot Cities, Chilled Economies* main report, `arshtrock2022hotcities`	Main report not obtained; the public methodology was	City figures are used as modelled context only, with the methodology as the source of the convention
ISO 15686-5:2017, `iso15686`	Paid standard; preview to clause 4.2.2 held	Used for life-cycle cost vocabulary; no service life, discount rate or escalation rate is attributed to it
ISO 7243:2017, `iso7243`	Paid standard; to be purchased before the comfort index is fixed	Fixes the instrument requirement in the equipment line, not any threshold value quoted here

A second and smaller class is cited from publication details alone, because the source is a standard, a classic text or a reference table used by definition rather than by quotation: `iso31000`, `iso22301`, `icc2020forcemajeure`, `bellman1957dp`, `powell2022rlso`, `howard1966voi`, `stewart2012lcz`, `jendritzky2012utci`, `beck2002words`, `w3c_provo` and `h3_restable`. They are listed here so that no reader mistakes their absence from an archive for an oversight. The thermal indices are the sharp case: the two documents that govern the comfort metric are both paid or paywalled, their threshold values must be read from the standards themselves, and buying them belongs in the pre-award phase rather than in the first work package.

A.3 Defects that make particular numbers unsafe to quote

Four defects are recorded because a reviewer who re-checks a figure and finds something different should know where the difficulty lies, and because each of them changes what this application is willing to claim.

Phoenix. The two cool-pavement reports are the strongest evidence in the register, and their city-wide water and energy extrapolations are the weakest part of them, because they scale a single day's measurement to a city. No such extrapolation is used here. Nor is any unit cost: neither report gives a service life, a re-application interval or a price, so no cost per square metre of coating may be taken from them, and the life-cycle cost of a coating remains unestablished. What the reports do support is graded measured-with-controls and points in one direction: road-centre mean radiant temperature in Phase II was 5.8°F higher at noon and 4.5°F higher in the afternoon on coated sections, a standing person needed 12 to 13 watts per square metre more heat loss, and the authors state that the road-centre measurement is not representative of a pedestrian on the adjacent pavement. Shade therefore ranks first in the candidate set, coatings are conditional, and the optimiser carries mean radiant temperature rather than air temperature.

Paris OASIS. The programme published a controlled before-and-after design with paired control stations and a documented traverse protocol, and published no effect sizes for the schoolyard transformations. It is used here for the design and never for a result. Its paving paper's albedo table and its microclimate paper's instrument-uncertainty table cannot be read reliably, which is a second reason no number is drawn from them.

Makkah and the regional comparators. The road-cooling and holy-site cooling figures reported in the region rest on agency statements relayed by the press, with no published protocol, no control and no instrument description. They are graded case-report, they establish that the interventions exist and are being scaled, and they are never used to support a claimed temperature reduction.

Megaproject statistics. The cost and schedule overrun literature describes projects two to four orders of magnitude larger than the [1.4m:389k](AED:USD) sought here. It is cited for one structural insight, that a schedule which cannot move pays for delay in a fixed window, and never as a calibrated uplift on this budget.

One further caution belongs to the gate rather than to a defect list. The Abu Dhabi study's seasonal tables carry displaced season labels, so only season-agnostic ranges are quoted from it: a root-mean-square error of 0.24–0.68°C for a model calibrated in one district and applied to another of the same class, and 0.27–1.62°C after bias correction (measured). Those are the units in which the month-12 gate's error condition is written, and they are the only published UAE precedent for it.

A.4 The evidence horizon

Every source registered here was secured and dated on 22 September 2026 by the companion feasibility study [Shastry et al., 2026], and that date is the evidence horizon of this application: nothing after it is claimed. Where a programme was still running at that date, the register records its position then.

Two consequences follow for anyone re-checking a figure. Announcement pages are revised in place, without notice and without a version history, so a policy-fact quoted here may not appear in the same words at the same address; that is why web sources carry a lower weight throughout, and why the call's own terms should be re-read against the call as published rather than as registered here. A formally issued document is stable in a way a page is not, which is why every measured result in this application rests on one.

What the register asks of this project

Fifty-one registered sources yield two with a contemporaneous untreated comparator, and both concern a surface treatment this application declines to rank first. The centre of gravity of the field is method-reference, policy-fact and case-report. That is the gap this work is designed to close: a paired-control campaign across one summer, with Zone 1 untreated throughout, would add the third such entry to the literature. The dataset and comfort maps that the month-12 gate makes the fallback deliverable will be registered in exactly this form, each series carrying its date of origin and its grade, so that a third party can audit them on the day the project closes.

1

B {B}

Identifier Schema and Data Dictionary

Abstract

This appendix is the field-level specification on which every measurement this application promises depends. It gives each key of the eight entities the corridor twin must store (place, sensor node, observation, intervention candidate, built intervention, scenario, decision and provenance) a name, a type, a definition, a worked example and a note saying what breaks if it is absent. One sticky flag, `is_synthetic`, separates records that may train the twin from records that may stand as evidence, and the provenance triple of entity, activity and agent makes that separation auditable. It closes with the five life-cycle indicators the schema makes computable and the one reference class this programme is in a position to create. It is a procurement document, not a description of one.

The quantities this application undertakes to report are not observed by any instrument. Restored Activity Hours, RAH, are hours joined to a place, an activity and a cohort, then differenced across an intervention version; cost per restored activity hour is that difference divided by a sum over cost elements; decision uptake is a brief joined to the agency that received it. Each is computed late and high in the architecture from records created early and low, and only the identifiers travel the whole distance. Identifiers inherit upward and cannot be added afterwards. A node shipped without a per-record calibration stamp yields a summer of readings that can never state which calibration they rest on, and no platform engineering recovers it. That is why the schema is frozen before the equipment order: the sensing line of [196k:54.4k](AED:USD), built up from unit rates in Section 2.5 (p. 12) (grade: derived), buys firmware behaviour as much as it buys hardware.

The second reason is that each of the four month-twelve gate conditions is a query over this dictionary and nothing else. Uptime is a count over `sensor_id` and `timeslot_start`; model error is a comparison keyed on `model_run_id` and `calibration_version`; the hour-by-hour comfort map of both zones is an aggregation over `place_id` and `timeslot_start`; and written confirmation of usefulness from a partner agency is a row bearing a `decision_id` (grade: design-target) [Iyyunni, 2026]. The dictionary is the gate in machine-readable form, and it belongs with the data-sharing requests rather than after them. It also governs the downside: if the gate fails, the programme closes with the dataset and platform as its deliverables, and a year of records keyed to place, time, sensor and calibration is then an asset in the sense of information

value [Howard, 1966], where the same year without those fields is a collection of numbers nobody can defend.

B.1 The field dictionary

Table B.1 (p. 26) is the dictionary. Spatial identity uses the H3 hexagonal indexing system at two resolutions: resolution 11, of about 2,150 m^2 average cell area and about 24.9 m average edge length, and resolution 13, of about 43.9 m^2 and about 3.6 m (grade: method-reference) *[Uber, 2025]*[1]. Resolution 11 carries the district field at 25–50 m, and so also the fused, physics-constrained 25–30 m city layer; resolution 13 carries the street field, which this application states as 1–5 m (grade: design-target) [Iyyunni, 2026]. The fit at the street scale is not exact, and the dictionary says so rather than leaving a supplier to discover it: a resolution-13 cell cannot represent structure below about 3 m, and a product claiming 1 m would have to be keyed at resolution 14, about 6.3 m^2 and 1.3 m (grade: method-reference) *[Uber, 2025]*[1]. Resolution-13 cells nest exactly inside resolution-11 cells, so a street result aggregates to the district layer without a spatial join. The place key is also the one field that must reconcile with a layer the programme does not control, since geometry is drawn from the Dubai Digital Twin Platform (grade: policy-fact) *[Dubai Media Office, 2026]*[2]. Time is a UTC instant with the local offset in a separate field, never folded into the timestamp, so that a policy window can be re-cut over the archive: the statutory midday break removes 12:30 to 15:00 from 15 June to 15 September (grade: policy-fact) *[The National, 2026]*[3], and reporting restored hours inside and outside it must not require the series to be rebuilt.

[1] Uber Technologies, *H3 Geospatial Indexing System: Table of Cell Areas and Edge Lengths by Resolution*, H3 documentation, 2025. https://h3geo.org/docs/core-library/restable

[2] Government of Dubai Media Office, *Hamdan bin Mohammed witnesses launch of Dubai Digital Twin Platform*, Dubai Media Office, 2 July 2026. https://mediaoffice.ae/en/news/2026/july/02-07/hamdan-bin-mohammed-witnesses-launch-of-dubai-digital

[3] The National, *Midday break for outdoor workers to begin on June 15*, 4 June 2026. https://www.thenationalnews.com/news/uae/2026/06/04/midday-break-for-outdoor-workers-to-begin-on-june-15/

Table B.1: Field dictionary for the corridor Decision Twin. Types are logical, not storage, types. Every field is mandatory at the layer that first writes it; a record missing one is rejected at ingestion rather than repaired downstream. Only the spatial scheme *[Uber, 2025]* and the provenance vocabulary [W3C, 2013] are taken from published standards; the remaining schemes are the project's own and are proposed here for agreement with the partner agencies before procurement.

Field	Type	Definition	Example	Notes
A. Place				
`place_id`	string	H3 cell index naming the ground area a record describes	`8b3f1a2c 4d65fff`	15 hexadecimal characters; the one key that must reconcile with a municipal layer *[Dubai Media Office, 2026]*
`place_res`	integer	H3 resolution; 11 for the district field, 13 for the street field	`11`	about 2,150 m^2 and 43.9 m^2 average cell area respectively *[Uber, 2025]*
`place_parent_id`	string	resolution-11 cell containing a resolution-13 cell	`8b3f1a2c 4d65fff`	written at ingestion, not inferred at query time
`place_zone`	enum	`zone1_heritage`, `zone2_corridor`	`zone2_corridor`	Zone 1 is untreated across all three summers and is the control
`place_class`	string	local climate zone label with the scheme version that produced it	`LCZ-proj-06/v1.2`	project composites, not the standard classes [Stewart and Oke, 2012, Manandhar et al., 2019]
B. Time				
`timeslot_start`	timestamp	UTC instant opening the bin	`2027-08-14 T09:30:00Z`	UTC only; a local instant is never stored in this field
`utc_offset`	string	the local offset in force at that instant	`+04:00`	kept separately so that policy windows can be re-cut over the archive
`bin_width_s`	integer	width of the stored bin, in seconds	`900`	nodes record on the ten-minute cadence [Iyyunni, 2026]; the store holds 900 s bins
`timeslot_rule`	string	versioned rule mapping node records into stored bins	`agg/v1.0/mean`	an analytical choice that changes values, so it is a field
C. Sensor node				
`sensor_id`	string	node or reference-station serial	`DXB-NODE-07`	nine nodes, two of them reference-grade, as scoped in Section 2.5 (p. 12)
`sensor_role`	enum	`case`, `control`, `reference`	`control`	a node without a role cannot support a paired comparison
`calibration_version`	string	semantic version with the date of calibration	`v2.1/ 2027-06-03`	stamped per record by firmware; unrecoverable if omitted

continued on the next page

Table B.1 (p. 26), continued from the previous page

Field	Type	Definition	Example	Notes
`firmware_version`	string	build identifier of the node software	`fw-1.4.2`	an unrecorded firmware change is invisible in the series
`variable`	enum	`air_temp`, `rel_hum`, `wind`, `rad`, `surf_temp`, `globe_temp`, `nat_wb`	`globe_temp`	the last two are the instrument addition the comfort index requires [ISO, 2017b, Jendritzky et al., 2012]
D. Observation				
`observation_id`	string	one variable, one node, one bin	`OBS/07/2027-08-14T09:30Z/air_temp`	a natural key, so that duplicates collide rather than accumulate
`value`	decimal	the quantity, in the unit declared beside it	`41.2`	never stored without its unit field
`unit`	enum	`degC`, `pct`, `m/s`, `W/m2`	`degC`	unit is a field, not a convention
`quality_flag`	enum	`ok`, `suspect`, `gap_filled`, `missing`	`gap_filled`	`gap_filled` counts against uptime and never towards it
E. Intervention candidate				
`candidate_id`	string	type, site and variant of an option in the optimiser	`SHADE/AGB-07/v1`	a candidate is a row in a solver, not a commitment
`candidate_type`	enum	`SHADE`, `TREE`, `PAVE`, `MIST`, `ROUTE`	`PAVE`	the objective carries mean radiant temperature, not air temperature [ASU and City of Phoenix, 2021, ASU and City of Phoenix, 2024]
`cost_element_id`	string	code in a breakdown structure of at least three levels	`1.2.3`	children sum to their parent, or the denominator is not auditable [GAO, 2020]
`perf_param`	decimal	the decaying property on which the candidate's effect depends	`0.36`	carries a re-measurement interval; reflectivity fell from 33–38 to 19–30 per cent within ten months (measured-with-controls) [ASU and City of Phoenix, 2021]
`service_life_yr`	decimal	assumed service life used in the life-cycle denominator	`15`	an assumption, printed in the same table as any result that uses it [ISO, 2017a]
F. Built intervention				

continued on the next page

Table B.1 (p. 26), continued from the previous page

Field	Type	Definition	Example	Notes
`build_id`	string	the one structure actually constructed	`BUILD/AGB-07/01`	joins the as-designed candidate to the as-built asset
`as_built_version`	string	monotone version distinguishing design from build	`v3`	supersedes without deleting, which is what the Year-3 comparison needs
`valid_from`, `valid_to`	timestamp	the period for which the record is asserted true	`2029-05-18T00:00:00Z`	an open `valid_to` means still in force
G. Scenario and model run				
`scenario_id`	string	weather year, demand path and price path	`WY2027/DEM-base/PR-high`	every twin output carries exactly one
`model_run_id`	string	model name, version and configuration hash	`envimet-4.x/cfg-9ab3d1`	the key against which any fidelity claim is reported
`fidelity_score`	decimal pair	error against a named held-out window, reported as a range	0.27–1.62°C	stated as the Abu Dhabi validation design states it (measured) [Manandhar et al., 2019]
H. Decision				
`decision_id`	string	brief, recipient, date and outcome	`BR-2028-04/RTA/adopted`	the gate's fourth condition is a count over this field
`decision_outcome`	enum	`delivered`, `acknowledged`, `adopted`, `declined`	`adopted`	delivered is not adopted, and the distinction is the impact measure
I. Provenance and generation				
`prov_entity`	string	the record itself, named as a PROV-O entity	`ent:obs/41a9`	every value the platform holds is an entity [W3C, 2013]
`prov_activity`	string	the activity that generated it	`act:ingest/v1.0`	linked by `wasGeneratedBy` [W3C, 2013]
`prov_agent`	string	the party responsible for that activity	`agt:bpdc/platform`	linked by `wasAttributedTo` [W3C, 2013]
`prov_derived_from`	string list	the entities this record was derived from	`[ent:obs/40f2]`	the closure of this field is the lineage [W3C, 2013]
`is_synthetic`	boolean	true if the record or any ancestor was generated	`true`	sticky under derivation; see Section B.2 (p. 30)
`generator_version`	string	program and version that produced a generated record	`tracegen/1.4.0`	a change of generator must be visible as a change of value
`seed`	integer	64-bit seed making a stochastic draw reproducible	`8814032117`	without it a scenario cannot be re-examined after challenge

continued on the next page

Table B.1 (p. 26), continued from the previous page

Field	Type	Definition	Example	Notes
`source_dataset_id`	string	the observed data a generator was fitted or conditioned on	`DS/summer-2027`	the field that makes leakage into a validation set detectable

Two rows carry more weight than their width suggests. `perf_param` exists because a cooling intervention's effect is not a constant: solar reflectivity measured at 33 to 38 per cent at installation had fallen to 19 to 30 per cent within ten months (grade: measured-with-controls) [ASU and City of Phoenix, 2021], so a performance figure recorded once misstates every ratio built on it. And `quality_flag` keeps gap filling honest: gaps will be filled, and the only defence against a filled value counting towards uptime is that it is marked where it is written.

B.2 The `is_synthetic` firewall

Every record in the store carries `is_synthetic`, observed records included, which carry it false. The flag is sticky: true if the record was generated by a model, a simulator or a statistical procedure, or if any ancestor was. Three rules follow, enforced as view definitions rather than stated as policy. Modelled and measured records are never mixed in one series: a comfort series built from observations and one built from downscaled fields are two series with two names, and they are never averaged. Any query that crosses the boundary must declare it, returning the proportion of contributing records that are synthetic, so that a brief cannot silently rest on an imputed ridership series. And no number offered as gate evidence, or as a figure a partner agency is asked to act on, may have a synthetic record anywhere in its lineage.

The firewall exists because all four gate conditions can be made to look satisfied without it. Uptime can be lifted by gap filling; model error can be made small by validating a field against the model that produced it; a comfort map can be complete because the holes were filled. None of that is fraud: each is the ordinary consequence of a convenient default, which is why the control belongs in the schema rather than in a reviewer's diligence. The rule does not forbid synthetic data, which is what a twin runs on. It is what makes it usable.

Retrospective keying

The single failure this appendix is written to prevent is a field added late. A key omitted at the sensing layer cannot be recovered at the decision layer, and a summer already collected cannot be re-keyed. The mitigation costs nothing: the dictionary is frozen before the equipment order, every change to it is versioned, and acceptance testing of the delivered nodes includes a check that each record carries every field of groups C and I of Table B.1 (p. 26).

B.3 Lineage

Lineage is recorded in the three terms of the W3C provenance vocabulary [W3C, 2013]. An *entity* is a thing with a value: a node reading, a bias-corrected field, a comfort map, a brief. An *activity* is what acted on entities to produce others: an ingestion run, a calibration, a model run, an optimisation. An *agent* is the party accountable, here the applicants' team for platform and decision work and the collaborating university for downscaling. Each derived entity points to its inputs, to the activity that produced it and to the agent. The lineage closure $L(r)$ is the transitive closure of those pointers, and the firewall is a predicate over it: a number is admissible as evidence only if no entity in $L(r)$ is flagged synthetic. Provenance is not documentation here. It is how a figure printed in a brief in 2030 is resolved, without the analyst who made it, back to the node, the calibration and the model configuration behind it.

B.4 What the identifiers make measurable

Five life-cycle indicators follow directly from the dictionary, and each names the one field it cannot survive without. Cost per restored activity hour, c_{RAH}, is the programme's single reporting unit: restored hours divided by discounted life-cycle cost, and it breaks without `cost_element_id` and `service_life_yr` [ISO, 2017a, GAO, 2020]. Sensor uptime is received records against expected records over `sensor_id` and `timeslot_start`, and it breaks without `quality_flag`. Calibration age is the interval between `timeslot_start` and the date inside `calibration_version`, and it is what tells a reader whether a late-summer reading was taken under a calibration three months old. Model error by place and by hour is `model_run_id` joined to held-out observations across `place_id` and `timeslot_start`, reported as a range in the manner of the Abu Dhabi study, whose errors were 0.27 to 1.62°C after bias correction (grade: measured) [Manandhar et al., 2019]. And decision uptake is a count over `decision_id` by `decision_outcome`, which is the only one of the five that no instrument can supply.

The reference class this work can create

Appraisal practice would price this programme against a reference class of comparable completed projects [Cantarelli et al., 2025]. No such class exists. No archived source in this application publishes a unit cost for a corridor-scale climate twin: not a cost per instrumented kilometre, not a cost per validated comfort-map hour, not a cost per brief delivered and adopted. The megaproject cost literature describes schemes orders of magnitude larger and must not be applied here as an uplift. The reference class this work most needs is therefore the one it is in a position to create, and it can do that only if its own costs are keyed to the same identifiers as its own results from the first month.

The cost of operating this schema has not been estimated. It falls in firmware, in platform engineering and in analyst discipline, and it belongs in the cost elements of Section 2.5 (p. 12) rather than being treated as free. Against it stands the cheapest insurance the programme can buy: agreeing, in writing and before procurement, what each field is called.

C {C}

Programme Detail

Abstract

The exhibits the main matter cites and has no room to print. Section C.1 (p. 32) sets the project against the six efforts that bound it. Section C.2 (p. 32) draws the layered architecture and the six things routinely treated as one. Section C.4 (p. 35) gives the fourteen milestones with the acceptance test that settles each. Section C.5 (p. 37) gives the risk register with a named owner on every line. Section C.6 (p. 40) builds the budget from unit prices and shows the two compliance tests it has to pass. Nothing here is asserted that is not used in the main matter, and nothing in the main matter rests on a figure that is not built here.

An application is read twice: once for the argument and once for the arithmetic. The main matter carries the argument in ten pages. This appendix carries the arithmetic, at the width the tables need.

C.1 The project against the global state of the art

Table C.1 (p. 33) is the evidence for Section 1.2 (p. 3). 'Loop closed' means a built intervention measured against a prediction registered before construction: the column exists because that is the test this project sets itself, and no row above it passes.

C.2 The layered architecture

Figure C.1 (p. 34) is the architecture behind Section 1.3 (p. 4). It marks six things routinely treated as one; each is a separate deliverable with its own owner and its own way of failing [Iyyunni, 2026].

C.3 The six institutional support measures

The call requires the endorsing institution to demonstrate six support measures, and an application that lists them without saying what each costs or who signs it has not demonstrated anything *[Dubai RDI Programme, 2025]*[1]. Each is answered here in the terms the institution will have to commit to.

[1] Dubai Research, Development and Innovation Program, *Research Projects Funding*, Dubai RDI programme page, 2025. https://dubairdi.ae/dubai-rdi-projects-funding/

Table C.1: This project against the global state of the art. 'Loop closed' means a built intervention measured against a prediction registered before construction. Grades: policy-fact for the programme facts, measured-with-controls for the Phoenix results, method-reference for the indices.

Effort	What it produces	Resolution	Loop closed?	What this project adds
Cooling Singapore *[Centre for Liveable Cities, 2024]*	Coupled urban climate simulation and scenario comparison for a tropical city	City, district	No built intervention measured against a prior twin prediction	An activity metric in place of an environmental state, and one intervention built and measured. OTC means outdoor thermal comfort there, Outdoor Tourism Capacity here
Destination Earth *[ECMWF, 2024]*	Global hourly digital twins of the Earth system	5–10 km; extremes twin 4.4 km globally, 500–750 m regionally	No	A street model at 1–5 m, two to three orders of magnitude below where those twins stop
Dubai Digital Twin Platform *[Dubai Media Office, 2026]*	City geometry: more than 195,000 buildings, 280,000 infrastructure assets, 1,500 layers; rainfall its named simulation	Asset, building	No	The hour-by-hour comfort layer it does not carry, and the brief that turns that layer into a choice
Paris OASIS schoolyards	A published controlled before-and-after design for schoolyard cooling; no effect sizes	Site; 1 m instruments	Partly: built and measured, with no prior prediction to test	The design imported intact, with the twin's prediction registered before the canopy is built
Phoenix cool pavement	Field trial of reflective coating against simultaneous untreated controls	Street segment	Partly: measured against controls, with no twin	Mean radiant temperature as the deciding quantity, which its own results require
Holiday Climate Index	Destination suitability from five variables in four weighted sub-indices	Daily; destination	No	Hourly and place-level, with a heat-stress override able to call a dry, calm Gulf afternoon unusable

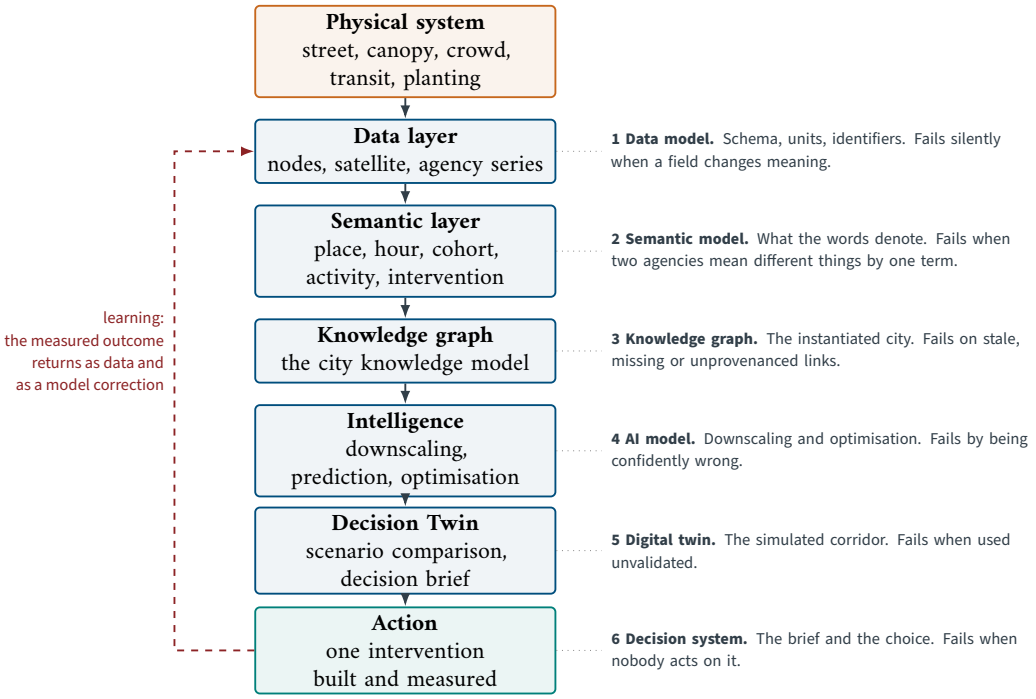


Figure C.1: The layered architecture, and the six things it keeps apart. Each is a separate deliverable with a named owner and a distinct failure mode; the dashed edge returns the measured outcome of the built intervention to the data layer and to the model that predicted it.

Reduced teaching load. The Letter of Institutional Commitment states the contact-hour reduction for the project lead and the four Co-Investigators. Five people at [2.4k:667](AED:USD) to [4.8k:1.33k](AED:USD) a month of released time for 36 months is [432k:120k](AED:USD) to [864k:240k](AED:USD) of institutional time (derived, on the market rates registered in the companion study [Shastry et al., 2026]). None of it is claimed from the grant, which is the point: it is the institution's own stake, and it is larger than the equipment line.

Fast-tracked procurement. The sensing line, [196k:54.4k](AED:USD), must reach purchase order within six weeks of grant start. The instrumentation has to be installed and settled before June 2027 or the first summer is lost, and a summer cannot be recovered inside a 36-month programme.

Expedited hiring. Both funded posts are advertised before award, not after it. The doctoral student starts in month one and the postdoctoral researcher in month three; the three-month offset is a deliberate saving taken in Section C.6 (p. 40) and not a recruitment delay.

Flexibility to recruit research staff. The two summer campaigns need a seasonal technician for fortnightly servicing through June to September, who is not one of the two funded posts. The institution must be able to engage one twice without opening a permanent position.

Letter of Institutional Commitment and *Letter of Collaboration*. The two written undertakings the call names by title. Neither is signed today, and Section 2.1 (p. 8) says so rather than implying otherwise; both are to be signed before submission.

C.4 Milestones and their acceptance tests

Table C.2 (p. 36) is the schedule behind Section 2.2 (p. 9). Each row states how completion is tested, because a milestone whose acceptance test is its own assertion cannot release money.

Table C.2: Milestones and their acceptance tests, 15 March 2027 to 14 March 2030. Dates, phases and the four gate conditions are the storyboard's and are design-targets rather than estimates [Iyyunni, 2026]; every acceptance test is the applicants' own. The error threshold in the validation row is set in the units of the one published UAE precedent [Manandhar et al., 2019].

Date	Phase	Deliverable	Acceptance test
Dec 2026	Pre-award	Site-access and data-access letters, Dubai Culture and Dubai Municipality	Signed letters naming the marked node positions, the geospatial layers released and a contact in each authority
Feb 2027	Pre-award	Data letters from the transport and utility authorities; collaboration agreement with Curtin University Dubai	Each letter names the dataset, its fields, its period and its release date; the agreement carries a non-zero, signed budget share
15 Mar 2027	Phase 1	Grant start; semantic layer and data dictionary version 1.0	Every entity carries an identifier, a unit and a provenance rule in the entity-activity-agent pattern [W3C, 2013]; `is_synthetic` declared on every record type
May 2027	Phase 1	Sensor network live	Nine nodes reporting on the ten-minute cadence for fourteen consecutive days, each with a calibration certificate and a `sensor_id` resolving in the store
Sep 2027	Phase 1	First summer dataset, June to September	Uptime computed per node across the season and published with every gap listed and attributed
Jan 2028	Phase 1	Model validated against the sensors	Bias-corrected hourly air temperature within 1.0°C root-mean-square error on a held-out week not used in the correction
Feb 2028	Phase 1	Hour-by-hour comfort map of both zones	Universal Thermal Climate Index and wet-bulb globe temperature for every hour of the season, as a dated file carrying its provenance
Mar 2028	Gate	Stop-or-go decision at month twelve	All four gate conditions evidenced in one pack; any one failing closes the project with the dataset and platform as deliverables
Nov 2028	Phase 2	MCN for the corridor; the intervention selected	A ranked candidate set carrying mean radiant temperature and not air temperature alone [ASU and City of Phoenix, 2024], with capital, energy, water and maintenance itemised, and the owning authority's written sustainment commitment attached to the selection
Feb 2029	Phase 2	Decision Twin version 1	A named officer at one partner runs a scenario comparison unaided and receives a brief of two pages or fewer; a three-dimensional rendering does not satisfy this row
May 2029	Phase 3	Intervention built over the most exposed 200 m	As-built survey lodged, and the twin's prediction for the treated segment time-stamped and lodged before construction begins
Sep 2029	Phase 3	Before-and-after comparison across one summer	Difference-in-differences against the untreated heritage-core control, reported against the recorded prediction whether or not the two agree
Feb 2030	Phase 3	Decision briefs to the four partners; transferable method document	Each brief names one decision, the option ranking, and the evidence grade of every quantity in it
14 Mar 2030	Close	Final report, dataset and publications	Dataset lodged under a persistent identifier; every quantity traceable to a graded source

C.5 The risk register

Table C.3 (p. 38) is the register behind Section 2.4 (p. 11), kept in the form of ISO 31000 with a named individual, not a committee, against every line [ISO, 2018].

Table C.3: Risk register with owners. Likelihood and impact are the applicants' own indicative assessment on a three-point scale and carry no evidence grade; the facts underneath are the call's rules *[Dubai RDI Programme, 2025]*, the storyboard's gate conditions [Iyyunni, 2026] and the companion study's costing [Shastry et al., 2026].

Risk	Likelihood	Impact	Mitigation	Owner
Consent and data letters do not arrive	Medium	High	One request per partner, each fitting on a page; corridor walk before the December letters; a signatory who is empowered and not merely senior; if only the heritage letter is late, the corridor is instrumented first	Principal Investigator
Written confirmation of usefulness not given (g_4)	Medium	High	Negotiated as a win condition before award; the comfort map shown in draft in December 2027, not February 2028; both authorities approached, so that either can satisfy the condition	Principal Investigator
Sensor uptime below 90 per cent across a Gulf summer	High	High	Dust, airborne salt and air temperatures above 45°C are the named failure modes and the enclosure specification is written against them; two spare nodes held; fortnightly service through the season; uptime published weekly	Co-Investigator, edge sensing
Model error above the threshold	Medium	High	The threshold is fixed in advance at 1.0°C on bias-corrected hourly air temperature, against the 0.27–1.62°C the one UAE precedent reached after bias correction (measured) [Manandhar et al., 2019]; if uncorrected error dominates, that is reported as the finding	Postdoctoral researcher, with Curtin University Dubai
Canopy consent refused or delayed by the heritage authority	Medium	Medium	The intervention site is the corridor and not the heritage core, precisely because the core admits no construction; a demountable, reversible structure is specified; consent sought on selection in November 2028, six months before the build; the fallback is a lighter option from the same optimiser run	Co-Investigator, built environment
Platform partner does not contract	Medium	Low	No platform is contracted in Year 1 and the Decision Twin interface is budgeted as in-house work in Year 2, so the partner is an option exercised in Phase 2 and not a dependency	Co-Investigator, systems integration
Scope exceeds the envelope	High	High	The descopes are taken in the open in Section C.6 (p. 40); the built structure and all operation after March 2030 are outside the grant and named as such; a scope change requires a change to Table C.4 (p. 40), not a reserve	Principal Investigator
Annual cap breached in Year 1	Medium	High	Year 1 is costed at exactly the cap and the equipment line is fixed-price before award; any addition displaces a node rather than the cap	Principal Investigator, with the institutional finance office
Loss of a funded post mid-term	Medium	High	Expedited hiring is among the institutional support measures committed to; calibration records, code and model configurations live in the institutional repository from month one, so that a replacement resumes rather than restarts	Principal Investigator

continued on next page

Table C.3: Risk register with owners (continued).

Risk	Likelihood	Impact	Mitigation	Owner
Data protection or privacy exposure in the pedestrian counts	Low	High	Counts are aggregated to the place-hour and arrive from the transport authority already aggregated; no image, device identifier or person-level key is retained, and the schema carries `place_id` and `time_slot` and nothing that resolves to an individual; the `is_synthetic` firewall keeps modelled records out of measured series, so a simulated count can never be reported as an observation [W3C, 2013]	Co-Investigator, systems integration, with the institutional ethics committee

C.6 The budget, built from unit prices

Table C.4 (p. 40) is the budget behind Section 2.5 (p. 12). The storyboard's budget section stated drafting directives, equipment 'reduced to' about [250k:69.4k](AED:USD) and computing to be 'raised to a realistic figure' [Iyyunni, 2026]; those are design-targets and these lines replace them, every figure a bottom-up estimate at 2026 prices, labelled derived.

Table C.4: The budget by category over the three project years, 2026 prices. Every line is derived: a bottom-up estimate at Dubai market rates, fully loaded for visa, medical cover, gratuity and employer costs, on the unit rates registered in the companion study [Shastry et al., 2026]. The annual totals row is the compliance test: no project year exceeds [500k:139k](AED:USD), and the three sum to [1.4m:389k](AED:USD) exactly. The dirham column is the binding arithmetic; each dollar equivalent is converted at the rounded working rate of 3.6 and rounded on its own, so a column of them can differ from its stated total by one cent.

Category	Year 1	Year 2	Year 3
Equipment: nine nodes with the wet-bulb globe correction, mounts, power, telemetry, spares	[196,000.00: 54,444.44] (AED:USD)	–	–
Personnel: postdoctoral researcher, months 3–36	[180,000.00: 50,000.00] (AED:USD)	[216,000.00: 60,000.00] (AED:USD)	[216,000.00: 60,000.00] (AED:USD)
Personnel: doctoral student, months 1–36	[96,000.00: 26,666.67] (AED:USD)	[96,000.00: 26,666.67] (AED:USD)	[96,000.00: 26,666.67] (AED:USD)
Collaborator share, Curtin University Dubai	–	[90,000.00: 25,000.00] (AED:USD)	[50,000.00: 13,888.89] (AED:USD)
Computing: two summers of street-scale simulation	–	[55,000.00: 15,277.78] (AED:USD)	[30,000.00: 8,333.33] (AED:USD)
Travel: field campaigns and partner meetings	[12,000.00: 3,333.33] (AED:USD)	[13,000.00: 3,611.11] (AED:USD)	[12,000.00: 3,333.33] (AED:USD)
Ethics review, data governance, publication	[16,000.00: 4,444.44] (AED:USD)	[10,000.00: 2,777.78] (AED:USD)	[16,000.00: 4,444.44] (AED:USD)
Annual total	**[500,000.00: 138,888.89] (AED:USD)**	**[480,000.00: 133,333.33] (AED:USD)**	**[420,000.00: 116,666.67] (AED:USD)**
Total, 36 months			**[1,400,000.00:388,888.89](AED:USD)**

The equipment line. Two reference-grade nodes at [40k:11.1k](AED:USD) each, seven calibrated nodes at [8k:2.22k](AED:USD) each, a black-globe and natural-wet-bulb assembly on each of those seven at [4.5k:1.25k](AED:USD), and [28.5k:7.92k](AED:USD) of mounts, power, telemetry and spares: [196k:54.4k](AED:USD) in total (derived). The seven assemblies, [31.5k:8.75k](AED:USD) of the line, are the item that cannot be cut, for the reason set out in Section 1.3 (p. 4): without them the corridor measures no index that a published labour series can be joined to [ISO, 2017b, Jendritzky et al., 2012].

Personnel and the collaborator share. [612k:170k](AED:USD) over 34 months for the postdoctoral post and [288k:80k](AED:USD) over 36 for the doctoral post; the collaborator share is [140k:38.9k](AED:USD), carried in Years 2 and 3.

The first compliance test: the annual cap. No project year may exceed [500k:139k](AED:USD) *[Dubai RDI Programme, 2025]*[1]. Year 1 is costed at exactly that, Year 2 at [480k:133k](AED:USD) and Year 3 at [420k:117k](AED:USD); the three sum to [1.4m:389k](AED:USD) (derived). An average year is not a year, which is why the table shows each: the ask divided by three gives [467k:130k](AED:USD), and a profile built on that average would have breached the cap in Year 1, when the instrumentation is bought.

The second compliance test: the Dubai share. At least 50 per cent of the grant must be utilised within Dubai *[Dubai RDI Programme, 2025]*[1]. The two personnel lines, [900k:250k](AED:USD), and the Curtin University Dubai share, [140k:38.9k](AED:USD), are spent in the emirate by residents of it: [1.04m:289k](AED:USD), or 74.3 per cent (derived). Procuring the instrumentation through a supplier registered in Dubai raises it to [1.24m:343k](AED:USD), or 88.3 per cent (derived).

The descopes, taken in the open. Four decisions make the scope fit the envelope, each taken rather than discovered: nine instrumented nodes rather than twenty, because nine that compute the index beat twenty that cannot; no field workstation; no contracted platform in Year 1; and the postdoctoral post beginning in month three rather than month one. Two things sit outside the grant: the built structure, and all operation, water, energy and repair after March 2030, both belonging to the owning authority under the life-cycle convention this programme reports in [ISO, 2017a]. Where ambition exceeds the envelope the response is a stated descope, not a declared reserve.

D {D}

The Simulation

Abstract

The model chain of Section 1.3 (p. 4) is not a proposal. It has been built and run end to end on a second Dubai corridor, 495 m of walking route in Dubai Academic City between The Myriad and the BITS Pilani Dubai Campus, and this appendix is the record of that dry run. Section D.1 (p. 42) states what it does and does not contain, which is prediction without measurement. Section D.2 (p. 43) gives the geometry and the standards it conforms to. Section D.3 (p. 44) gives the route from geometry to a comfort field. Section D.4 (p. 44) gives the predicted benefit of one canopy. Section D.5 (p. 45) reports a published canopy specification that failed its own test under this model, and the correction. Section D.6 (p. 46) reports how far the reanalysis can be trusted against real instruments. Section D.7 (p. 47) ranks what moves the answer, by measurement rather than by intuition. Section D.8 (p. 47) states what none of it licenses. The whole appendix is graded modelled and uncalibrated at the site, and no figure in it is carried to the intervention corridor.

A reviewer scoring the methodology criterion is entitled to ask whether the model chain exists or is being promised. It exists. Between the Letter of Intent and this application the chain was implemented and run, not on the intervention corridor, where no instrument may lawfully be placed until the access letters conclude, but on a corridor the team can reach on foot: the walk from The Myriad to the BITS Pilani Dubai Campus in Dubai Academic City, about 20 km inland from the heritage core. Eleven pipeline steps run from source download to a diurnal comfort animation. What that buys this application is narrow and worth stating plainly: it converts the software risk in Section 2.4 (p. 11) into a known quantity, and it means the grant buys instruments, access and validation rather than a first implementation.

It buys nothing else. The dry run has no site measurement in it anywhere, which is the subject of Section D.1 (p. 42) and the reason every figure below carries the same grade.

D.1 What the dry run contains, and what it does not

The study is organised on the same four steps the main matter uses, measurement, prediction, validation and decision, and it is honest about stopping after the second. Table D.1 (p. 43) is the status of each.

Three consequences follow and are carried through this appendix rather than noted and

Table D.1: Status of the Academic City dry run against its own four-step protocol. Step 2 is built; step 1 is not, and steps 3 to 5 are design only. The asymmetry is the point: what this application asks to fund is the half that is missing.

Step	Status	What stands in its place today
1 Measure	**Not started**, design only	ERA5 reanalysis from a grid cell 1.81 km east of the corridor. Route geometry is the only measured quantity in hand, and building heights are assumed for 131 of 132 footprints
2 Predict exposure	**Done**, three routes	Sky view factor, mean radiant temperature, physiological equivalent temperature (PET) and the Universal Thermal Climate Index at 124 to 176 points per route
3 Human response	Instrument drafted, no data	A survey instrument to ISO 10551 and ASHRAE 55, held for ethics approval. No human data has been collected and none should be inferred
4 Predict benefit	**Done**, one canopy	Two canopy geometries modelled over the route at three design conditions
5 Validate, then decide	Design only	Pass criteria agreed in advance; the computational fluid dynamics (CFD) cases are written and have **not been run**

forgotten. Every atmospheric quantity is reanalysis, so no statement here is an observation. Pedestrian wind comes from a one-dimensional canopy-layer profile rather than from the CFD cases, so it is the weakest term in the chain and is known to be. And because no local calibration of PET exists, PET is reported as a physical quantity and as a difference and is never translated into a verdict such as comfortable: the published assessment bands were derived for Central European subjects, and applying them to an acclimatised Gulf population would overstate the discomfort of the baseline and overstate the benefit of the intervention at the same time.

D.2 The corridor and its geometry

The walkway network and building footprints are OpenStreetMap, under the Open Database Licence *[OpenStreetMap, 2025]*[1]; the meteorology is ERA5 through Open-Meteo *[Open-Meteo, 2025]*[2], which is Copernicus Climate Change Service information [Hersbach et al., 2020]. All analysis is in EPSG:32640, which is WGS 84 UTM zone 40N and therefore in metres, with interchange in CRS84; the container is an OGC GeoPackage of eight layers [OGC, 2021], the interchange format is RFC 7946 GeoJSON [Butler et al., 2016], and the meteorological series carry CF-1.10 standard names and SI units. The Overpass query and

[1] OpenStreetMap contributors, *Copyright and Licence*, OpenStreetMap Foundation, 2025. Walkway network and building footprints are © OpenStreetMap contributors and are licensed under the Open Database Licence (ODbL) 1.0. https://www.openstreetmap.org/copyright

[2] Open-Meteo, *Historical Weather API: ERA5 reanalysis*, API documentation, 2025. ERA5 is Copernicus Climate Change Service information; the service is licensed CC-BY 4.0. https://open-meteo.com/en/docs/historical-weather-api

the Open-Meteo request are recorded verbatim so that the inputs can be regenerated rather than trusted.

Twenty-nine walkways totalling 8.5 km were captured and each scored for its relation to the main route, which gives the canopy study a defensible candidate set rather than an arbitrary one. Each centreline is refitted as an arc-length-parameterised cubic spline, which is what supplies continuous tangents for orienting a canopy module and curvature for flagging where a rigid module will not follow the path.

Three routes carry the comparison, and the pair it rests on is not the obvious one. Route A is 496 m and Google-derived. Route B, 487 m, is the OpenStreetMap shortest path and is *not* an alternative itinerary: it is the same physical corridor, 100 per cent of it within 25 m of route A at a mean separation of 8.6 m, and it functions as a geometry-uncertainty twin. That it returns nearly identical exposure is a consistency check and not a finding, and it is reported as such. Route C, 699 m with 83 per cent of it beyond 25 m from route A, is the genuine second itinerary.

D.3 From geometry to a comfort field

Sky view factor is computed by ray casting against extruded building footprints. Mean radiant temperature is assembled from the six radiant flux terms, with ground and canopy surface temperatures solved from an energy balance rather than assumed, which matters here because the unshaded paving solves to about 82 °C at the design hour and an assumed value would have carried the whole result. PET follows Höppe's Munich energy balance model [Höppe, 1999] through the `pythermalcomfort` implementation [Tartarini and Schiavon, 2020] rather than a re-derivation, and the Universal Thermal Climate Index is reported beside it [Jendritzky et al., 2012].

Two body geometries are carried throughout, and the distinction is what makes step 5 testable. One is a standing person, six-directionally weighted with the beam through a vertical projected area, and it feeds PET. The other is a sphere, which is what a globe thermometer actually reads, and it exists so that a measured globe temperature can later be compared with a modelled quantity of the same shape. They differ by about 8 K on this corridor, because the paving is hot and a sphere sees far more of it than a standing body does. A validation campaign that compared a globe reading against the standing-person figure would find an 8 K discrepancy that is an artefact of geometry and not an error, and the pass criteria in Table D.1 (p. 43) are written against the sphere for that reason.

One defect found in the dry run is worth recording because it is the kind that does not announce itself. Pairing radiation to the hour label rather than to the hour centre produced 422 spurious sunlit hours across the series. It was found in step 3 and corrected to zero. An instrumented campaign logging to Coordinated Universal Time with local time derived, as Section 1.3 (p. 4) specifies, does not have this failure mode, and the specification is written that way because of this.

D.4 The predicted benefit of one canopy

The modelled intervention is a pinnate compound leaf: a structural spine on the centreline carrying leaflets that alternate left and right, each independently tiltable and separately addressed, over the 2.7432 m walking path. On route A that is 331 bays, 662

leaflets and 84 masts at an amortised pitch of 1.4956 m, so that modules close out exactly on the spline with no stub bay. A leaflet tapering to a point cannot tile a rectangular band, and the chord-to-pitch ratio that fixes it follows in closed form at 1.442; plan coverage is then measured by polygon union rather than assumed, at 0.9975.

Table D.2 (p. 45) gives the result at three design conditions.

Table D.2: Predicted canopy benefit on route A of the Academic City corridor, at three design conditions. Grade: modelled, and uncalibrated at the site. PET is a physical quantity here and a difference, never a comfort verdict. These figures belong to this corridor and this geometry and are not carried to the intervention corridor (Section D.8 (p. 47)).

Design condition	Baseline PET	With canopy	ΔPET	Route in direct beam
Peak heat, 46.3 °C	62.5 °C	48.9 °C	−13.6 K	100 % to 0 %
Typical hot, 41.3 °C	56.8 °C	43.8 °C	−12.9 K	100 % to 0 %
Shoulder, 27.7 °C	39.7 °C	36.2 °C	−3.5 K	100 % to **85 %**

Key finding

The shoulder-season row is the design finding, and it is not a disappointment. At a solar elevation of 37 degrees the sun passes beneath a 2.74 m canopy and 85 per cent of the route stays in direct beam, against nothing at all at peak heat. A 9 ft width is sized for high sun. If shoulder-season performance matters, the answer is not a wider canopy but the actuators, and that is a control problem rather than a structural one. An optimiser scored only on the peak-heat row would never have found this.

Figure D.1 (p. 46) is one frame of the diurnal field, at noon on the hot-season 95th-percentile day of the reanalysis record. The benefit tracks solar elevation almost exactly across the day, from −3.5 K at 08:00 to −14.6 K at noon and back to −2.2 K at 18:00, for a daylight mean of −8.5 K. The 16:00 profile is the one that repays attention: at low sun the protection becomes a comb, alternating along the route according to how the local path bearing sits against the sun azimuth, which is a geometric effect no route-mean figure would reveal.

D.5 A specification that did not survive its own test

A second canopy geometry was modelled, and the reason to report it here is not the geometry but what happened when it was tested. The design is a four-pole tensile hyperbolic paraboloid bay, 8 m wide on a 15 m span, with pole heights derived from the solar profile angle, and it claims an equinox switch: deep shade through the summer, and admission of the low winter sun for warmth. Its solar geometry was verified independently against `pvlib` [Holmgren et al., 2018] and agrees to within 0.37 degrees, so the astronomy is sound.

The switch is not. As specified, the design takes a minimum over two conditions that pull in opposite directions, because the cap needs the lowest summer sun while the floor needs the highest winter sun. Run on the real route, it holds 73.4 per cent of the corridor shaded in December at −12.0 K, which is to say it delivers summer shade in winter and the

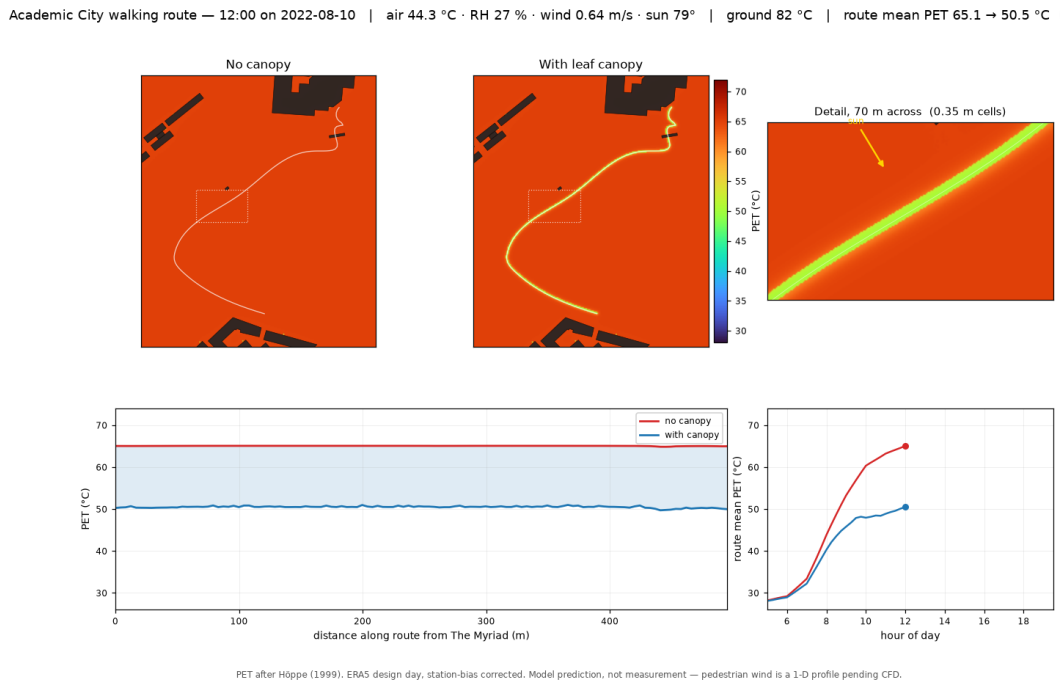


Figure D.1: Modelled comfort field on the Academic City corridor at noon on a hot-season design day, with and without the leaf canopy, with a 0.35 m detail inset and the along-route profile. Grade: modelled, and uncalibrated at the site. The figure carries its own caption to the same effect, because a frame lifted out of a slide deck should not be able to lose it: pedestrian wind is a one-dimensional profile and the CFD has not been run.

promised admission does not appear. Solving the switch objective directly instead, rather than deriving it, recovers 4.57 K of winter warmth for 0.36 K of summer shading, and 8.87 K at December noon.

A second limit is structural and sets the two geometries against each other usefully. An 8 m bay on a bend requires a radius of at least 76 m before its distortion exceeds the design's own 10 per cent flag, and only 70.7 per cent of route A provides it; 9 of 34 bays exceed the flag. Where it fails hardest is the last 70 m into the campus, which is precisely the stretch a 1.5 m-pitch leaf follows without difficulty. The defensible scheme is both geometries, chosen by curvature, and that is a conclusion neither design would have reached on its own.

Two things are being demonstrated by including this section, and neither is the canopy. The first is that the chain is sharp enough to falsify a published specification on geometry the authors did not choose, which is the property Section 1.3 (p. 4) needs it to have before it can test the twin's own prior prediction in Year 3. The second is the habit: a specification was tested, it failed, the failure is in the record with its cause, and the correction is reported beside it.

D.6 How far a reanalysis can be trusted

The honest weakness of the dry run is that it has no instrument at the corridor. What it does have is a comparison against real instruments 15 km away, at Dubai International, across 35,064 overlapping hours. This establishes a regional reanalysis bias

and emphatically not site conditions, and Table D.3 (p. 47) reports it in the pool that matters, hot-season daylight hours.

Table D.3: ERA5 against instrumented observations at Dubai International, WMO 41194, 15 km from the corridor, over 35,064 overlapping hours. Grade: measured for the station series, modelled for ERA5. The hot-season daylight pool, 6,990 hours, is the one the design conditions are drawn from. This is a regional bias and not a site measurement, and it does not retire the instrumentation that Section 1.3 (p. 4) specifies.

Variable	Bias, ERA5 minus station	RMSE	Correlation, all hours
Air temperature	+0.85 K	1.95 K	0.955
Relative humidity	−8.5 points	12.6 points	0.833
Wind speed at 10 m	−0.12 m s^{-1}	1.57 m s^{-1}	0.712

The reanalysis runs warm and dry in the hot-season daylight pool. Correcting for it moves absolute PET by about 2 K but moves ΔPET by only 0.46 K, which is the single most useful number in this appendix: the difference between shaded and unshaded is robust to a bias that the absolute value is not. It is why Section 1.3 (p. 4) validates against a difference, why the Year-3 design is paired case and control rather than a single reference station, and why the gate threshold is set on bias-corrected air temperature rather than on raw.

Wind is the term least well served, at a correlation of 0.712 against an instrument, and the pedestrian wind beneath the canopy is worse still because it comes from a one-dimensional profile over a reanalysis wind 1.81 km away. The OpenFOAM cases that would replace it are written, georeferenced back to EPSG:32640 and configured for three wind sectors [Weller et al., 1998], and they have not been run. Section 2.5 (p. 12) carries the computing line that runs them.

D.7 What actually moves the answer

Table D.4 (p. 48) ranks the assumptions by measured effect on ΔPET rather than by intuition, and the ranking is not the one the study began with.

The result worth carrying forward is the first row. Two years of effort could be spent reducing geometric uncertainty, and the measured answer is that it would buy about a third of a kelvin at the design hour, while a procurement decision about the reflectance of a membrane is worth six times that. This is the value-of-information argument of Section 1.1 (p. 2) arriving as a number on a real corridor [Howard, 1966], and it is the kind of finding the instrumented campaign exists to make at Al Ghubaiba, where the ranking will be different and cannot be guessed from here.

D.8 What the dry run licenses, and what it does not

This section is the firewall, and it is the one part of the appendix a reader in a hurry should not skip.

Table D.4: Measured sensitivity of ΔPET to each assumption, swept one at a time at the peak-heat condition unless stated. Grade: modelled. The ranking is by measured effect and is deliberately not the ranking the study expected.

Assumption	Swept over	Moves ΔPET	What follows
Canopy skin reflectance	0.30 to 0.85	2.2 K	The largest sensitivity found, and it is a specification decision rather than a modelling problem. A canopy is procured with a reflectance, and this is the line on which that procurement should be argued
Ground albedo	0.15 to 0.40	0.8 K	Second, and it is the surface the optimiser is also choosing, which couples two candidate interventions that are easy to treat as independent
Reanalysis bias	raw to station-corrected	0.46 K	Absolute PET moves about 2 K on the same correction. The difference is the robust quantity, and the decision rests on it
Building height	6 to 25 m	0.36 K at peak heat, 0.52 K at low sun	The assumption the study most expected to matter, and it matters least, because the design-case sun is nearly overhead. It is still the first thing a survey should retire, because it sets the baseline sky view factor as an upper bound

What is not claimed here

No figure in this appendix is a prediction for the Year-3 intervention. The Academic City corridor is about 20 km inland from the heritage core, on open campus ground rather than in narrow lanes between coral-stone walls, at a baseline sky view factor of 0.98 against a shaded historic fabric that is already passively cooled. The canopy modelled here is not the canopy that will be built, the design conditions are not those of the Bur Dubai corridor, and the exposure the intervention has to remove is a different quantity. The front matter states that this application forecasts no number of degrees and no number of restored hours for the Year-3 canopy, because the size of that effect is what the experiment exists to measure and the twin's prior prediction of it is the object of the test rather than an input to it. That statement stands unqualified. A −13.6 K from Academic City is evidence that a model chain runs, and it is evidence of nothing whatever about Al Ghubaiba.

Three limits bound what the dry run can support even on its own corridor, and all three are the reasons this application is for instruments rather than for software. Building heights are assumed for 131 of 132 footprints, so the baseline sky view factor of 0.98 is an upper bound and should be read as one. No local calibration of PET exists, so absolute PET bands do not apply and only the difference carries a decision. And the CFD has not been run, so pedestrian wind is unvalidated. None of the three blocks a single pilot bay. All three block a 500 m commitment, and the dry run says so in its own protocol rather than being told so by a reviewer.

What the dry run does license is a claim about readiness, and it is a narrow one worth stating exactly. The eleven pipeline steps from source download to comfort field exist, run, and are reproducible from recorded queries. The geodetic and interchange standards are conformed to rather than intended. The sensitivity ranking is measured rather than asserted. A published specification has been tested against the chain and found wanting on its own terms, which is the capability Section 1.3 (p. 4) requires before a prior prediction can be registered and then tested in Year 3. On the evidence of this appendix the methodology criterion is not being asked to take a model chain on trust, and the twelve-month gate is not being asked to deliver a first implementation under field conditions.

The one thing the dry run cannot give itself is the thing the grant buys. Every atmospheric quantity in it comes from a reanalysis grid cell 1.81 km from a corridor nobody has instrumented. Nine nodes on the Al Ghubaiba corridor, each carrying the black globe and the natural wet bulb that Section 1.3 (p. 4) adds to the specification, turn every modelled figure in this appendix into a quantity that can be found wrong.

Glossary

The glossary is grouped by subject rather than run together as a single alphabet, so that a reviewer who knows one domain but not another can find a term by the family it belongs to: heat and comfort, cities, data, decisions, money, delivery and institutions. Within each of the seven groups the entries are alphabetical, and acronyms are listed as cross-references to the spelt-out term rather than defined twice. Terms are defined as this application uses them; where a term carries a wider meaning elsewhere, the narrower sense intended here is stated, and where a definition rests on an argument made in the body, the section is named. Page numbers for these definitions appear in bold in the Index.

G.1 Climate and thermal comfort

Bias correction The adjustment of a model's systematic departure from observations, fitted on a period for which both exist. It is distinct from calibration, which changes parameters, and from validation, which uses data the model has not seen. The one regional precedent reports street-scale errors of 0.27–1.62°C after correction (grade: measured), the order of magnitude in which Section 1.3 (p. 4) sets the gate's threshold.

Cool pavement A coating of raised solar reflectivity applied to a road or a footway. The controlled desert trial in the archive found treated streets about 12.0°F cooler at noon but the mean radiant temperature at the road centre about 5.5°F higher (grade: measured-with-controls), which is why Section 1.2 (p. 3) ranks shade first and treats coatings as conditional.

Downscaling The inference of a field at fine spatial resolution from observations and model output at coarser resolution. Physics-informed downscaling constrains that inference by the governing equations rather than by statistical fit alone, and is the machine-learning task named in Section 1.3 (p. 4).

Local climate zone A classification of urban land into types that share a thermal behaviour, defined by building height and density, surface cover and materials rather than by land use. It lets a model calibrated on one district serve another of the same class, a transfer the regional precedent tested at 0.24–0.68°C (grade: measured).

Mean radiant temperature The single temperature of an imagined uniform enclosure that would exchange the same radiant heat with a person as the real surroundings do. It is how shade and surface reflectivity act on a body, and the reason a reflective pavement can cool a street while making a pedestrian standing on it hotter.

Physiological equivalent temperature The air temperature of a reference indoor setting

in which a person's heat balance would be the same as in the outdoor setting being assessed, computed from a two-node energy balance. Appendix D (p. 42) reports it as a physical quantity and as a difference and never as a verdict such as comfortable, because the published assessment bands were derived for Central European subjects and a population acclimatised to a hot arid climate reports neutrality at substantially higher values. The difference between shaded and unshaded is the robust quantity; the absolute value is not.

Reanalysis A gridded reconstruction of past weather produced by assimilating historical observations into a fixed modern forecasting model, which makes it physically consistent and continuous in a way an observational record is not. It is not an observation. Every atmospheric quantity in Appendix D (p. 42) is reanalysis from a cell 1.81 km from the corridor, and that is the gap the instrumentation in Section 1.3 (p. 4) exists to close.

Sky view factor The fraction of the sky hemisphere visible from a point, between nothing at all and unity in the open. It governs how much long-wave radiation a surface can lose to the sky and how much direct beam reaches a person, and it is the single geometric quantity through which a canopy, a building and a street width all act.

Universal Thermal Climate Index A physiological index of the outdoor thermal environment as a person experiences it, expressed as an equivalent temperature. This application adopts it for pedestrians and reports wet-bulb globe temperature beside it, so that one hour can be read for a visitor and for a worker (Section 1.3 (p. 4)).

Wet-bulb globe temperature An occupational heat-stress index combining natural wet-bulb, black-globe and air temperatures in fixed proportions under a numbered international standard, and the scale in which labour ministries set work and rest rules. It requires three instruments and not one, which is the equipment correction Section 2.5 (p. 12) costs.

LCZ See *Local climate zone*.

MRT See *Mean radiant temperature*.

PET See *Physiological equivalent temperature*.

SVF See *Sky view factor*.

UTCI See *Universal Thermal Climate Index*.

WBGT See *Wet-bulb globe temperature*.

G.2 Urban systems, mobility and tourism

Heat-Lost Activity Hours The activity that would have taken place at a place and an hour and does not, because the thermal environment is outside the acceptable range for the people who would have been there. It is reported in activity-hours and not in money, because no archived source values a visitor-hour (Section 1.1 (p. 2)).

Minimum Cooling Network The smallest set of shade, planting, material and rest-point interventions that brings priority routes and spaces to a comfort target, chosen at

least cost across capital, energy, water and maintenance. It is the optimiser's output in Section 1.3 (p. 4), and deliberately not a list of everything that could be built.

Outdoor Tourism Capacity How many visitors can comfortably use a place, for how long and at which hours, on a given kind of day: physical carrying capacity weighted by the share of each hour in which the thermal environment is acceptable to the cohort using it. Note the collision of abbreviation: in the Singapore literature the same three letters mean outdoor thermal comfort.

Paired control A treated place and an untreated place of the same kind, measured in the same minutes, so that a difference between them cannot be attributed to the weather of the day. It is what separates a measured-with-controls result from a measured one, and the reason the heritage core is instrumented but never treated (Section 1.3 (p. 4)).

Restored Activity Hours The activity-hours an intervention returns to a place, computed as the reduction in Heat-Lost Activity Hours it produces. It is the numerator of this application's efficiency ratios, reported per dirham, per litre of water and per kilowatt-hour (Section 2.5 (p. 12)).

Shade canopy A built structure that interrupts direct solar radiation over a route or a space. The archived evidence ranks it first among candidate interventions because it acts on mean radiant temperature, which is the pathway that governs what a person feels.

HLAH See *Heat-Lost Activity Hours*.

MCN See *Minimum Cooling Network*.

OTC See *Outdoor Tourism Capacity*.

RAH See *Restored Activity Hours*.

G.3 Data, semantics and synthetic data

Data model What is recorded: the tables, fields, units and identifiers in which observations are held. It can be complete while the meaning of its fields remains unstated, which is the condition the semantic layer exists to remedy, and it is the first of the six layers this application keeps distinct.

H3 cell A cell of a global hexagonal spatial index, at one of a fixed series of resolutions with published cell areas. Resolution 11, of average edge about 24.9 m, carries the district layer and resolution 13, at about 3.6 m, the street layer (grade: method-reference), so the two aggregate without re-projection (Appendix B (p. 24)).

Is-synthetic flag The field `is_synthetic`, carried by every record, stating whether its content was observed or generated by a model. It is the firewall that keeps modelled values, gap-filled and bias-corrected ones among them, out of measured series, and a query that crosses the boundary must declare it (Appendix B (p. 24)).

Knowledge graph The linked instances of a domain, each with its provenance: this place, this canopy, this hour. It is populated from the data layer through the semantic layer and is distinct from both, and from the models, the twin and the decision system above it.

Provenance The record of what an item of data is, what produced it and who was responsible, expressed as entity, activity and agent. It is what allows a figure in a decision brief to be traced back to an instrument reading and a calibration date.

Semantic layer The statement of what each recorded field means and how the concepts relate to one another: heat exposure, pedestrian, destination, shade, road, building, energy demand, vulnerability, intervention. It is the first thing to be written, and the layer whose absence is the commonest cause of a system that retrieves but cannot reason.

G.4 Decision science and modelling

Decision system The layer in which alternatives are compared against a stated objective and a person decides. It is the last of the six layers and the only one whose output is an action; a twin that is queried but decides nothing has stopped one layer short (Section 2.3 (p. 9)).

Decision Twin The digital twin and the decision system taken together: an environment that compares candidate interventions before they are built and reports the comparison in a form a decision maker can act on. Section 1.4 (p. 6) places it above the emirate's existing geospatial platform rather than beside it as a competitor.

Digital twin An assembled, updated representation of a physical system that can answer questions of the form 'what if'. It is not a three-dimensional rendering, and it is not the geometry on which it sits.

Life-Cycle Architecture anchor point The stakeholder commitment milestone at which the architecture and the whole-life plan are reviewed together and the level of resource commitment changes. Life-Cycle Objectives precedes it, where scope and win conditions are agreed, and Initial Operational Capability follows, where a system first enters operational use; Section 2.3 (p. 9) maps the month-twelve review onto the middle one.

Spiral model A development process in which a system is built in successive loops, each resolving the largest remaining risk and each ending at a commitment milestone where the work may be stopped. Its incremental-commitment principle is the reason funds are released in two parts rather than one.

Stochastic dynamic programming The formulation of a sequence of decisions under uncertainty as a state, a set of decisions, exogenous information, a transition and an objective, solved backwards through time. It is the frame in which the observation year's value is the information it buys rather than the data it collects.

Value of information The improvement in expected outcome obtainable by observing something before deciding, less what the observation costs. An observation that would change no decision has a value of information of zero, however interesting it is, and that is the test every proposed sensor must pass.

LCA See *Life-Cycle Architecture anchor point*.

SDP See *Stochastic dynamic programming*.

VoI See *Value of information*.

G.5 Economics, costs and finance

Life-cycle cost The whole cost of an asset or a service over its life: acquisition, operation, maintenance, renewal and disposal, assessed over a stated period of analysis, and not the purchase price. On the one comparator with published operating costs, ownership over a service life exceeds construction several times over.

Reference class forecasting Estimating a project's outcome from the distribution of outcomes of a class of similar completed projects, rather than from a build-up of its own parts. Its discipline is choosing a class that genuinely contains the project: overrun statistics drawn from megaprojects describe work orders of magnitude larger than a research grant and are not applied to one here.

Total cost of ownership The life-cycle cost of a digital system in particular: licences, hosting, storage, support, upgrades and the eventual migration or retirement, priced for the whole term rather than for the first year. These items, and not the capital works, decide whether a capability survives the funded period.

LCC See *Life-cycle cost*.

RCF See *Reference class forecasting*.

TCO See *Total cost of ownership*.

G.6 Programme management and risk

Design-target A figure that states an intention rather than an estimate: the number an author was instructed to write, or chose to write, before any costing existed behind it. It sits outside the seven evidence grades because it is not evidence; the storyboard's equipment line of about [250k:69.4k](AED:USD) is the principal case, and Section 2.5 (p. 12) replaces such figures with costed lines and says so.

Evidence grade The standing of the source behind a quantity, in the seven-grade scheme this application inherits and applies: measured-with-controls, measured, modelled, projection, policy-fact, case-report and method-reference. A figure takes the grade of its weakest supporting link, this document's own arithmetic is labelled derived or indicative, and Appendix A (p. 16) registers the grade of every source used.

Stage gate A point at which a programme is reviewed against pre-agreed conditions and may be stopped, with a costed fallback defined in advance: an option to abandon, written into the schedule. The month-twelve review is one, requiring among its four conditions a sensor network live at 90 per cent uptime or better across the first summer (design-target) and an hour-by-hour comfort map of both zones; its fallback is that the dataset and the platform stand as the deliverables (Section 2.2 (p. 9)).

G.7 Institutions and programmes

Cognitive Cities The funder's priority domain, covering 'Smart mobility', 'Next-generation built environments and infrastructure', 'Climate-neutral and resilient

urban systems' and 'Urban health intelligence for proactive well-being'. Section 1.4 (p. 6) states which of the four topics this application serves.

Cooling Singapore The research programme that built a digital urban climate twin for a hot, humid city state. It is the closest comparator for the modelling stack, the benchmark Section 1.2 (p. 3) measures against, and the source of the collision on the three letters this application uses for outdoor tourism capacity.

Dubai Digital Twin Platform The emirate's geospatial platform, launched on 2 July 2026 with three-dimensional models of more than 195,000 buildings and 280,000 infrastructure assets across more than 1,500 geospatial layers, rainfall being its named simulation (grade: policy-fact). It supplies the geometry on which a decision layer can sit and is not a competitor to it (Section 1.4 (p. 6)).

Dubai Electricity and Water Authority The emirate's electricity and water utility, asked in this application for one dataset: district cooling load for the testbed district, hourly, for one summer.

Dubai Future Foundation RDI Grant The research, development and innovation grant programme to which this application is made. Its published rules cap the annual budget per project at [500k:139k](AED:USD) and require at least 50 per cent of the grant to be utilised within Dubai (grade: policy-fact); its ten evaluation criteria order the sections of the main matter.

Phoenix cool pavement A municipal reflective-coating trial that measured treated streets against adjacent untreated ones on the same day, and reported both the surface cooling and the increase in radiant load at the road centre. It is the strongest measured-with-controls evidence in the archive and the basis for ranking shade ahead of coatings.

Roads and Transport Authority The emirate's roads and transport authority, asked in this application for one dataset: station exit counts with bus and abra ridership for the testbed corridor.

DEWA See *Dubai Electricity and Water Authority*.

DFF See *Dubai Future Foundation RDI Grant*.

DUCT See *Cooling Singapore*.

RDI See *Dubai Future Foundation RDI Grant*.

RTA See *Roads and Transport Authority*.

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